

9

Gradient-Based Lithographic Optimization: Using the Normalized Image Log-Slope

There are many approaches for optimizing a lithographic process. Very commonly, the focus–exposure matrix is used to measure depth of focus (which includes a sensitivity to both dose and focus errors) and the process is optimized to maximize the depth of focus for select critical features. Such experimental optimization efforts are most effective and efficient when guided by appropriate theory. Here, the gradient-based approach to lithographic optimization, using the well-known image log-slope, is presented. By calculating and then optimizing the gradients of the image that is projected and recorded into a photoresist through focus, a clear direction for maximizing depth of focus is seen. This approach can also be thought of as a first-order propagation of errors analysis – a Taylor series approach to how input variations affect the output of CD. Small errors in exposure dose affect CD depending on the focus-dependent slope of the image at the nominal line edge.

9.1 Lithography as Information Transfer

This chapter will focus on CD control using exposure latitude and the variation of exposure latitude through focus as the basic metrics of CD control. The approach taken here will view lithography as a sequence of information transfer steps (Figure 9.1). A designer lays out a desired pattern in the form of simple polygon shapes. This layout data drive a mask writer so that the information of the layout becomes a spatial variation of transmittance (chrome and glass, for example) of the photomask. The information of the layout has been transferred, though not perfectly, into the transmittance distribution of the mask. Next, the mask is used in a projection imaging tool to create an aerial image of the mask.

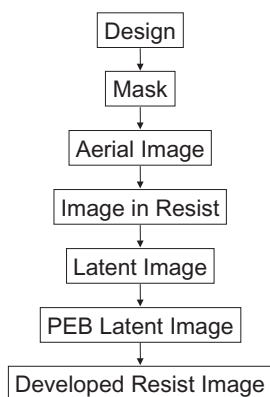


Figure 9.1 *The lithography process expressed as a sequence of information transfer steps*

However, due to the diffraction limitations of the wavelength and lens numerical aperture, the information transmitted to the wafer is reduced (see Chapter 2). The aerial image is an imperfect representation of the information on the mask. Aberrations, flare and other nonidealities degrade the image quality further (see Chapter 3). This aerial image is projected into the resist where the variation of transmittance with angle, resist refraction and substrate reflection makes the image in resist different than the aerial image (see Chapter 4). Exposure then produces a latent image of exposed and unexposed sensitizer concentration, which now contains some portion of the information originally stored as an aerial image (see Chapter 5). Post-exposure bake changes this latent image into a new latent image through diffusion (for a conventional resist) or reaction and diffusion (for a chemically amplified resist), with the result being a loss of information in the latent image (see Chapter 6). Finally, development converts the latent image into an image of development rates, which then is transformed into the final resist profile via the development path (see Chapter 7).

At each stage of this sequence, there is potential for information loss. The final developed resist image is not a perfect representation of the original design data. By defining metrics of information content at each stage of this sequence of information transfer steps, a systematic approach toward maximizing the information contained in the final image can be found. The information contained in each image propagates through the lithography process, from the formation of an aerial image $I(x)$ and image in resist, to the exposure of a photoresist by that image to form a latent image of chemical species $m(x)$ or $h(x)$, to post-exposure bake where diffusion and possibly reactions create a new latent image $m^*(x)$, and finally to development where the latent image produces a development rate image $r(x)$ that results in the definition of the feature edge (a one-dimensional example is used throughout this chapter for simplicity).

9.2 Aerial Image

Projection imaging tools, such as scanners, steppers, or step-and-scan tools, project an image of a mask pattern into air, and then ultimately into the photoresist. The projected

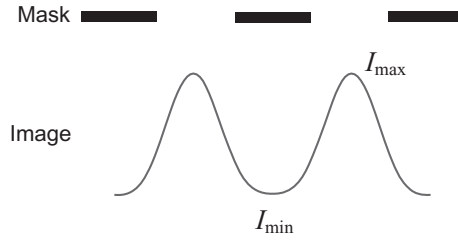


Figure 9.2 Image contrast is the conventional metric of image quality used in photography and other imaging applications, but is not directly related to lithographic quality

image in air is called the *aerial image*, a distribution of light intensity as a function of spatial position within (or near) the image plane. The aerial image is the source of the information that is exposed into the resist, forming a gradient in dissolution rates that enables the three-dimensional resist image to appear during development. The quality of the aerial image dictates the amount of information provided to the resist, and subsequently the quality and controllability of the final resist profile.

How do we judge the quality of an aerial image? If, for example, aerial images are known for two different values of the partial coherence, how do we objectively judge which is better? Historically, the problem of image evaluation has long been addressed for applications such as photography. The classical metric of image quality is the *image contrast* (Figure 9.2). Given a mask pattern of equal lines and spaces, the image contrast is defined by first determining the maximum light intensity (in the center of the image of the space) and the minimum light intensity (in the center of the line) and calculating the contrast as

$$\text{Image Contrast} = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} \quad (9.1)$$

Since the goal is to create a clearly discernible bright/dark pattern, ideally $I_{\min}$ should be much smaller than $I_{\max}$, giving a contrast approaching 1.0 for a high-quality ('high-contrast') image.

Although this metric of image quality is clear and intuitive, it suffers from some problems when applied to lithographic images. First of all, the metric is only strictly defined for equal lines and spaces. Although it is possible to modify the definition of image contrast to apply, for example, to an isolated line or to a contact hole, it is not clear that these modified definitions are useful or comparable to each other. It is clearly impossible to apply this metric to an isolated edge. Secondly, the image contrast is only useful for patterns near the resolution limit. For large features, the image contrast is essentially 1.0, regardless of the image quality. Finally, and most importantly, the image contrast is not directly related to practical metrics of lithographic quality, such as resist linewidth control.

Fundamentally, the image contrast metric samples the aerial image at the wrong place. The center of the space and the center of the line are not the important regions of the image to worry about. What is important is the shape of the image near the nominal line edge. The edge between bright and dark determines the position of the resulting

photoresist edge. This transition from bright to dark within the image is the source of the information as to where the photoresist edge should be. The steeper the intensity transition, the better the edge definition of the image, and as a result the better the edge definition of the resist pattern. If the lithographic property of concern is the control of the photoresist linewidth (i.e. the position of the resist edges), then the image metric that affects this lithographic result is the slope of the aerial image intensity near the desired photoresist edge.

The slope of the image intensity I as of function of position x (dI/dx) measures the steepness of the image in the transition from bright to dark. However, to be useful it must be properly normalized. For example, if one simply doubles the intensity of the light, the slope will also double, but the image quality will not be improved. (This is because intensity is a reciprocal part of exposure dose, which is adjusted to correctly set the final feature size in photoresist.) Dividing the slope by the intensity will normalize out this effect. The resulting metric is called the image log-slope (ILS):

$$ILS = \frac{1}{I} \frac{dI}{dx} = \frac{d \ln(I)}{dx} \quad (9.2)$$

where this log-slope is measured at the nominal (desired) line edge (Figure 9.3). Note that a relative change in image intensity is the same as a relative change in exposure dose, so that the ILS is also a relative exposure energy gradient. Since variations in the photoresist edge positions (linewidths) are typically expressed as a percentage of the nominal linewidth, the position coordinate x can also be normalized by multiplying the log-slope by the nominal linewidth w , to give the normalized image log-slope (NILS).

$$NILS = w \frac{d \ln(I)}{dx} \quad (9.3)$$

The NILS is the best single metric to judge the lithographic usefulness of an aerial image.¹⁻³

Since the NILS is a measure of image quality, it can be used to investigate how optical parameters affect image quality. One of the most obvious examples is defocus, where the NILS decreases as the image goes out of focus. Figure 9.4a shows the aerial image of a space at best focus, and at two levels of defocus. The ‘blurred’ images obviously have a lower ILS at the nominal line edge compared to the in-focus image. By plotting the log-slope or the NILS as a function of defocus, one can quantify the degradation in aerial

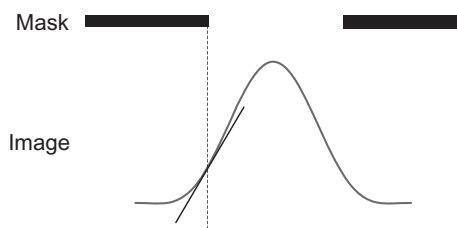


Figure 9.3 Image Log-Slope (or the Normalized Image Log-Slope, NILS) is the best single metric of image quality for lithographic applications

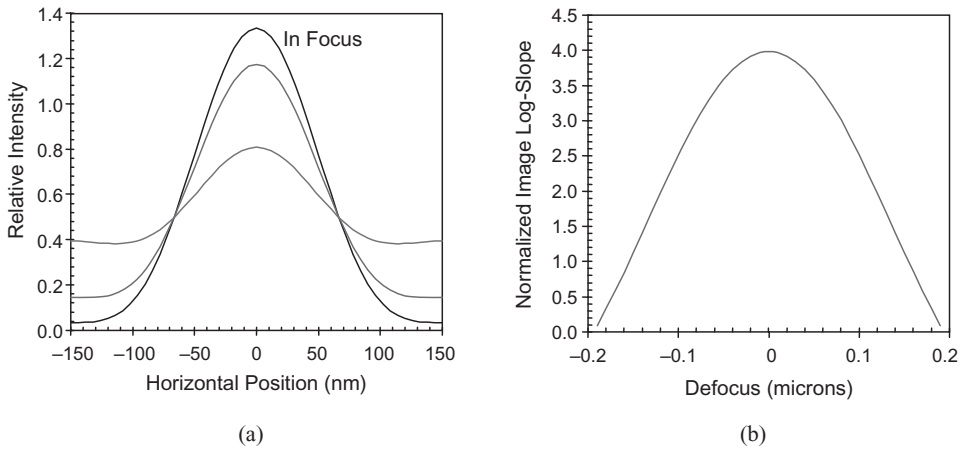


Figure 9.4 The effect of defocus is to (a) ‘blur’ an aerial image, resulting in (b) reduced log-slope as the image goes out of focus (150-nm space on a 300-nm pitch, $NA = 0.93$, $\lambda = 193\text{ nm}$)

image quality as a function of defocus (Figure 9.4b). This log-slope defocus curve provides a very important tool for understanding how focus affects a lithographic process. For example, suppose one assumes that there is a minimum acceptable NILS value, below which the aerial image is not good enough to provide adequate resist images or linewidth control. In Figure 9.4b, for example, a minimum acceptable NILS value of 1.5 would mean that this imaging process can tolerate about $\pm 0.14\text{ }\mu\text{m}$ of defocus and still get aerial images of acceptable quality. Thus, an estimate of the minimum acceptable NILS can lead to an estimate of the depth of focus. Note that, in general, the focus response of an aerial image is symmetric about best focus. When this is the case, the log-slope defocus curve is usually plotted only for positive values of defocus.

To see how the log-slope defocus curve can be used to understand imaging, consider the effects of wavelength and numerical aperture on the focus behavior of an aerial image. Figure 9.5a shows how the NILS of a $0.25\text{-}\mu\text{m}$ line/space pattern degrades with defocus for three different wavelengths (365, 248 and 193 nm), all other lithographic parameters held constant. It is clear from the plot that the lower wavelength provides better image quality for the useful range of defocus. For a given minimum acceptable value of NILS, the lower wavelength will allow acceptable performance over a wider range of focus. One could conclude that, for a given feature being imaged, a shorter wavelength provides better in-focus performance and better depth of focus.

The impact of numerical aperture (NA) is a bit more complicated, as evidenced in Figure 9.5b. Here, the log-slope defocus curves for three different numerical apertures (again, for a $0.25\text{-}\mu\text{m}$ line/space pattern) cross each other. If one picks some minimum acceptable NILS value, there will be an optimum NA which gives the maximum depth of focus (for example, a minimum NILS value of 2.5 has the best depth of focus when $NA = 0.6$). Using a numerical aperture above or below this optimum reduces the depth of focus.

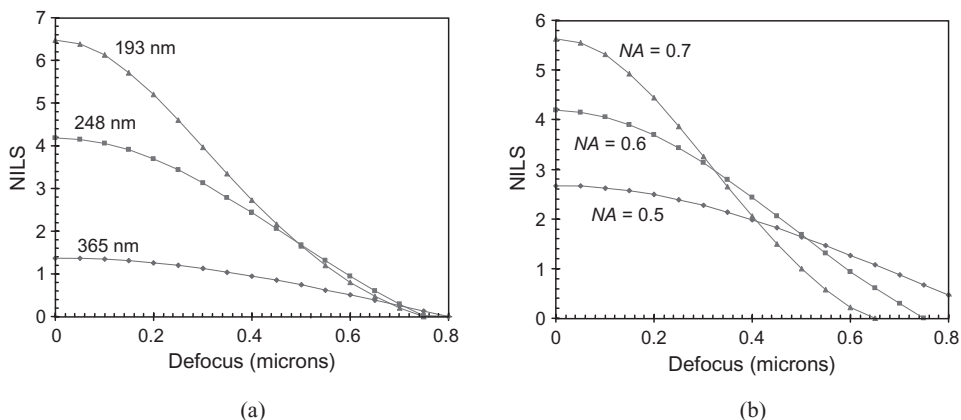


Figure 9.5 Using the log-slope defocus curve to study lithography: (a) lower wavelengths give better depth of focus ($NA = 0.6$, $\sigma = 0.5$, 250-nm lines and spaces), and (b) there is an optimum NA for maximizing depth of focus ($\lambda = 248\text{nm}$, $\sigma = 0.5$, 250-nm lines and spaces)

NILS values are easy and fast to calculate and provide a simple yet valuable metric of image quality. As an example of using this metric, the log-slope defocus curve is one of the easiest ways to quantify the impact of defocus on image quality. By using this tool, we have quickly arrived at two fundamental imaging relationships: when imaging a given mask pattern, (1) lower wavelengths give better depth of focus, and (2) there is an optimum NA that maximizes the depth of focus. But to make the best use of the NILS as an image metric, one must relate the NILS numerical value to lithographically measurable quantities. How does one determine the minimum acceptable NILS? If the NILS is increased from 2.0 to 2.5, what is the lithographic impact? More fundamentally, why is NILS a good image metric?

The answers to these questions lie with the fact that NILS is directly related to the printed feature's exposure latitude (as will be shown in subsequent sections of this chapter). Exposure latitude describes the change in exposure dose that results in a given change in linewidth. Mathematically, it can be expressed as one over the slope of a critical dimension (CD) versus exposure dose (E) curve, $\partial CD / \partial E$. Since both CD and dose variations are most usefully expressed as a percentage of their nominal values, the more useful expression of exposure latitude is $\partial \ln CD / \partial \ln E$. For the simplifying case of a perfect, infinite contrast photoresist the exposure latitude can be related to NILS by

$$\frac{\partial \ln E}{\partial \ln CD} = \frac{1}{2} \text{NILS} \quad (9.4)$$

To put this in more familiar terms, if we define exposure latitude to be the range of exposure, as a percentage of the nominal exposure dose, that keeps the resulting feature width within $\pm 10\%$ of the nominal size, then for an infinite contrast resist the exposure latitude can be approximately related to NILS by

$$\% \text{Exposure Latitude} \approx 10 * \text{NILS} \quad (9.5)$$

(the approximation comes from the assumption that NILS is constant over the $\pm 10\%$ CD range). Thus, in a perfect world (i.e. a perfect photoresist), the impact of NILS can be easily related to a lithographically useful metric: each unit increase in NILS gives us 10 % more exposure latitude. Unfortunately, the real world is not so perfect and infinite contrast photoresists have yet to enter the commercial market. The real impact of NILS on exposure latitude is somewhat reduced from the above ideal. In general, Equation (9.5) can be modified to account for the nonideal nature of photoresists to become

$$\% \text{Exposure Latitude} \approx \alpha(\text{NILS} - \beta) \quad (9.6)$$

where α and β are empirically determined constants. β can be interpreted as the minimum NILS required to get an acceptable image in photoresist to appear. α then is the added exposure latitude for each unit increase in NILS above the lower limit β . By the end of this chapter, a more accurate and physically meaningful relationship between NILS and exposure latitude will be derived.

The values of α and β can be determined by comparing a calculated NILS-versus-defocus curve to experimentally measured exposure latitude-versus-defocus data. Figure 9.6 shows a simulation of such an experiment for a typical case. Once calibrated, a minimum acceptable exposure latitude specification (say, 10 %) can be translated directly into a minimum acceptable NILS value (in this case, 1.7). Since α and β are resist and process dependent, the minimum acceptable NILS must be also. And of course, the requirements for the minimum acceptable exposure latitude will impact the required NILS directly. Thus, either using Equation (9.5) for the ideal case, or Equation (9.6) for a calibrated resist case, a quantitative valuation of the importance of NILS can readily be made.

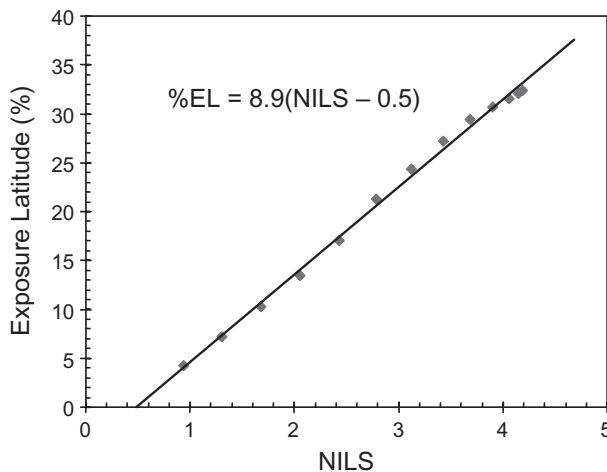


Figure 9.6 Typical correlation between NILS and exposure latitude (simulated data, $\lambda = 248\text{nm}$, $\text{NA} = 0.6$, $\sigma = 0.5$, 500nm of UV6 on ARC on silicon, printing 250-nm lines and spaces through focus)

The NILS is a generic metric of aerial image quality. Thus, it can be used to optimize any parameter that affects image quality. Some typical use cases include:

- Optimizing numerical aperture (NA) and partial coherence (σ)
- Investigating off-axis illumination parameters
- Comparing phase-shifting mask designs
- Understanding the impact of optical proximity correction (OPC) on image quality and optimizing OPC
- Examining the impact of aberrations, polarization and flare.

As an example, consider the first use case, optimizing NA and σ (the so-called optimum stepper problem). Most steppers and scanners made today have software-controlled variable numerical aperture and variable partial coherence. For a given feature or features, the goal is to pick the settings that maximize lithographic quality. Using the NILS as a proxy for lithographic quality, there are two basic optimization approaches: (1) for a given level of defocus, find the settings that produce the maximum NILS, or (2) for a given minimum acceptable NILS (that is, a given minimum allowed exposure latitude), find the settings that maximize depth of focus.

Figure 9.7 shows an example where the defocus is set to equal the built-in focus errors of the process ($0.2\mu\text{m}$ in this case), and contours of constant NILS are plotted as a function of NA and σ ($\lambda = 248\text{nm}$ printing 130-nm lines on a 360-nm pitch). The maximum NILS of 1.24 is achieved when $NA = 0.72$ and $\sigma = 0.72$. This optimum is sensitive to the amount of defocus used. If the defocus requirement can be lowered to $0.16\mu\text{m}$, the optimum stepper settings for this problem become $NA = 0.76$ and $\sigma = 0.64$ producing a maximum NILS of 1.43.

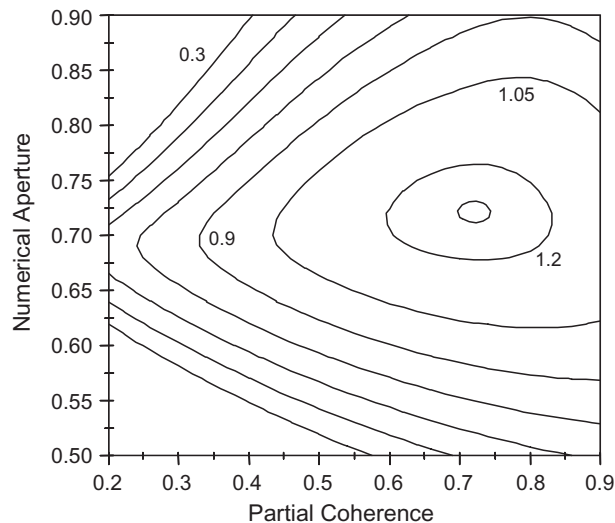


Figure 9.7 One approach to the optimum stepper problem is to pick a fixed amount of defocus ($0.2\mu\text{m}$) and find the settings that maximize the NILS ($\lambda = 248\text{nm}$, 130-nm lines on a 360-nm pitch, contours of constant NILS)

9.3 Image in Resist

As described in Chapter 4, the image in resist will differ from the aerial image due to the different efficiencies of coupling plane wave energy into the resist as a function of angle and polarization. For a reflective substrate, each incident angle will produce a different reflectivity swing curve, so that for a given resist thickness each incident angle will have a different transmittance into the photoresist. For a nonreflective substrate, the transmittance of the air–resist interface creates an angular and polarization dependence. Figure 9.8 illustrates these two cases.

The electric field aerial image can be thought of as the sum of planes waves, each with magnitude a_n , combining to form the image. For the case of TE-polarization,

$$E(x, z) = \sum_{n=-N}^N a_n e^{ik_x \sin \theta_n} e^{ik_z \cos \theta_n} \quad (9.7)$$

where z is the distance from the plane of best focus, θ_n represents the angle of the n th diffraction order with respect to the optical axis, and $\pm N$ diffraction orders are used to form the image. The image in resist can be calculated in the same way, but with the magnitude of each plane wave modified by the influence of the photoresist (see Chapter 4):

$$a_{nr} \equiv a_n \frac{\tau_{12}(\theta_n)}{1 + \rho_{12}(\theta_n) \rho_{23}(\theta_n) \tau_D^2(\theta_n)} e^{-\alpha_{\text{eff}} z/2} \quad (9.8)$$

where θ_n now represents the angle of the n th diffraction order in resist, α_{eff} is the effective absorption coefficient, as defined in Chapter 4, and z is the depth into the resist. For the

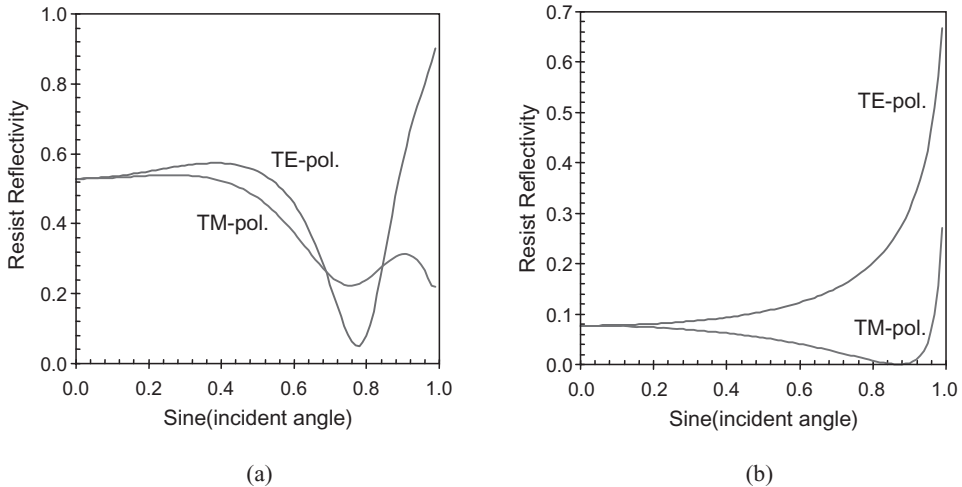


Figure 9.8 Intensity reflectivity between air and resist of plane waves as a function of incident angle and polarization for (a) resist on silicon, and (b) resist on an optically matched substrate ($\lambda = 248 \text{ nm}$, resist $n = 1.768 + i0.009868$)

case when a reasonably good BARC is used ($|\rho_{23}| \ll 1$), the impact of the film stack is dominated by the transmittance of the air–resist interface and absorption in the resist:

$$a_{nr} \approx a_n \tau_{12}(\theta_n) e^{-\alpha z / (2 \cos \theta_n)} \quad (9.9)$$

Once the impact of the resist film is properly accounted for and the image in resist has been calculated, the image log-slope concept can readily be applied to the image in resist. As before, the image-log slope and the NILS, evaluated at the nominal resist edge for the image in resist, will serve as the metric of information content of the image in resist. Additionally, defocus of the image as it propagates through the thickness of the resist means that the location of the image must also be specified. In general, the bottom of the resist will be used for calculating NILS, though sometimes the middle of the resist will prove more convenient.

9.4 Exposure

The NILS is, fundamentally, an aerial image or image in resist metric. It can be thought of as a measure of the amount of information contained in the image that defines the proper position of the desired photoresist edge. The information in this image propagates through exposure into a latent image of exposed and unexposed resist, where a similar metric must be defined to judge the quality of the latent image. Consider a simple yet common case: a resist with first-order kinetics (almost always the case) whose optical properties do not change with exposure dose (commonly the case for chemically amplified resists). For such a case, the latent image $m(x,y,z)$ is related to the intensity in the resist $I_r(x,y,z)$ by

$$m(x,y,z) = \exp(-CI_r(x,y,z)t) \quad (9.10)$$

where C is the exposure rate constant of the resist, t is the exposure time, and m is the relative concentration of light-sensitive resist material (see Chapter 5).

Equation (9.10) is the exposure image transfer function, translating an image in resist into a latent image. From our experience with using the NILS, one would expect that a slope or gradient of the latent image would serve as a good metric of latent image quality. The slope of the latent image (at the nominal feature edge position, for example) can easily be derived by taking the derivative of Equation (9.10), giving (for a simple 1D case)⁴

$$\frac{\partial m}{\partial x} = m \ln(m) \frac{\partial \ln I}{\partial x} \quad (9.11)$$

Thus, the latent image gradient is directly proportional to the image log-slope (and thus the normalized latent image gradient is proportional to NILS). This entirely logical result is very satisfying, since it means that all efforts to improve the NILS will result directly in an improved latent image gradient.

Equation (9.11) reveals another important factor in latent image quality. The term $m \ln(m)$, which relates the image log-slope to the latent image gradient, is exposure dependent (m being the relative amount of resist sensitizer that has not been exposed at

the point where the latent image gradient is being described). A simple thought experiment reveals the necessity for a dose dependence to latent image quality. At the two extremes of zero dose and infinite dose, the final latent image becomes a uniform chemical distribution [$m(x,y,z) = 1$ for zero dose, and $m(x,y,z) = 0$ for infinite dose]. Thus, the gradient of the latent image is zero at these extremes and no information is transferred to the wafer. But obviously the gradient is nonzero at doses in between these extremes. By the *Mean Value Theorem* of calculus, there must be a dose which maximizes the latent image gradient.

A plot of $-m \ln(m)$ versus m shows that there is one exposure dose (one value of m) that will maximize the latent image quality (Figure 9.9). When $m = e^{-1} \approx 0.37$, the value of $-m \ln(m)$ reaches its maximum and the full information of the aerial image is transferred into the resist during exposure. It is interesting to note that when $m = 1$ (no exposure) and $m = 0$ (complete exposure of the resist), the latent image gradient is zero and no information is transferred from the aerial image into the resist, as expected.

There are many interesting implications that come from the simple observation of the existence of an optimum exposure dose. Often, dose is used as a 'dimension dial', adjusting dose to obtain the desired feature size without regard to any process latitude implications. If the dose is near the optimum, this approach is valid. If, however, the dose used is significantly off from optimum (say, very underexposed compared to the peak of Figure 9.9), changing dose will affect both dimension and overall latent image quality.

The above discussion assumes a resist that does not bleach. Since g-line and i-line resist materials generally exhibit a large amount of bleaching, some attempts to describe how bleaching affects the latent image gradient will be useful. In general, the combined effects of bleaching with the propagation of the image from top to bottom of the resist yield a very complex picture. However, a few approximations can greatly simplify this picture without greatly sacrificing the accuracy of the derived latent image gradient.

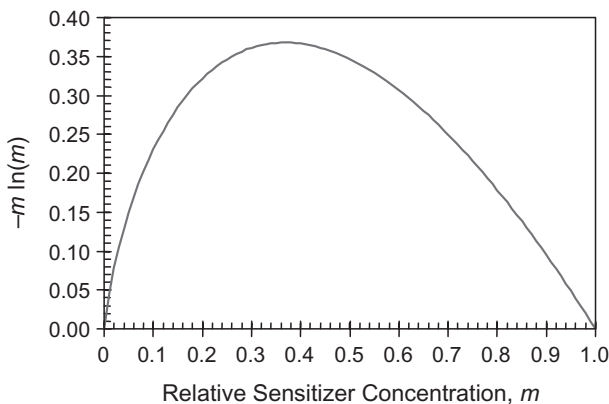


Figure 9.9 Plot revealing the existence of an optimum exposure, the value of m at which the latent image gradient is maximized. Note that $m = 1$ corresponds with unexposed resist, while $m = 0$ is completely exposed resist

When only absorption affects how the intensity varies with depth into a bleaching resist (that is, when there are no standing waves and when the defocusing of the image through the resist is ignored), Lambert's law of absorption coupled with Beer's law gives

$$\frac{dI}{dz} = -(Am + B)I \quad (9.12)$$

The coupled Equations (9.10) and (9.12) were solved in Chapter 5 to give

$$z = \int_{m(z=0)}^{m(z)} \frac{dy}{y[A(1-y) - B \ln(y)]} \quad (9.13)$$

where y is a dummy variable for the purposes of integration, and

$$I(x, z) = I(x, 0) \frac{A[1 - m(x, z)] - B \ln[m(x, z)]}{A[1 - m(x, 0)] - B \ln[m(x, 0)]} \quad (9.14)$$

Differentiating Equation (9.13) with respect to x gives the interesting result that

$$\frac{\partial m}{\partial x} = \left[\frac{m(x, z)I(x, z)}{m(x, 0)I(x, 0)} \right] \frac{\partial m}{\partial x} \bigg|_{z=0} \quad (9.15)$$

But at the top of the resist, bleaching does not affect the solution to the kinetic rate equation so that Equations (9.10) and (9.11) still apply at the top of the resist. Thus,

$$\frac{\partial m}{\partial x} = m(x, z) \ln[m(x, z)] \frac{\partial \ln I}{\partial x} \bigg|_{z=0} \quad (9.16)$$

This derivation shows that Equation (9.16) coupled with Equation (9.13) is a more general result for the latent image gradient after exposure and that Equation (9.11) is a special case of (9.16) when $z = 0$ or when $A = 0$ (since the image log-slope is not a function of z when there is no bleaching).

Calculating the gradient of the latent image when bleaching is occurring still involves the numerical integration of Equation (9.13). However, for the special case of $B = 0$ an analytical solution is possible. As was shown in Chapter 5,

$$m(x, z) = \frac{m(x, 0)}{m(x, 0) + [1 - m(x, 0)]e^{-Az}} \quad \text{when } B = 0 \quad (9.17)$$

which allows Equation (9.16) to be written as

$$\begin{aligned} \frac{\partial m}{\partial x} &= \left[\frac{m(x, 0) \ln[m(x, 0)] e^{-Az}}{(m(x, 0) + [1 - m(x, 0)]e^{-Az})^2} \right] \frac{\partial \ln I}{\partial x} \bigg|_{z=0} \\ &= \left[\frac{e^{-Az}}{(m(x, 0) + [1 - m(x, 0)]e^{-Az})^2} \right] \frac{\partial m}{\partial x} \bigg|_{z=0} \end{aligned} \quad (9.18)$$

By plotting Equation (9.18), Figure 9.10 shows that bleaching results in an increased latent image gradient. Like the contrast enhancement layers discussed in Chapter 4, bleaching allows greater transmittance of light in the bright spaces relative to the edges

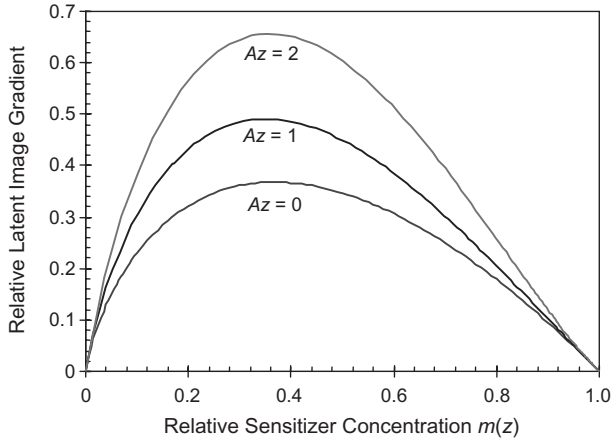


Figure 9.10 Bleaching (increasing values of Az) results in increased latent image gradient at the bottom of the resist (shown here is the special case where $B = 0$)

between lines and spaces, which necessarily have a lower image intensity. As a result, the light near the edge of the feature is more highly absorbed than light in the center of the space. The transmitted image $I(x,z)$ will have a higher slope than the image at the top of the resist. Interestingly, the optimum level of exposure remains very close to the non-bleaching case: $m(x,z) \approx e^{-1}$.

9.5 Post-exposure Bake

The post-exposure bake takes the latent image of exposure products and creates a new latent image at the end of the bake, $m^*(x)$. For a conventional resist, diffusion will spread out the latent image. For a chemically amplified resist, diffusion is accompanied by reaction to form a new latent image.

9.5.1 Diffusion in Conventional Resists

Fickian diffusion, where the diffusivity of the diffusing material remains constant, can be treated as a simple convolution problem (as described in Chapter 5). If $m(x)$ is the original latent image (the spatial distribution of chemical species in the resist), the new latent image after diffusion $m^*(x)$ can be found by convolving the original chemical distribution with a Gaussian sometimes referred to as the diffusion point spread function (DPSF):

$$m^*(x) = m(x) \otimes \text{DPSF} = \frac{1}{\sqrt{2\pi}\sigma_D} \int_{-\infty}^{\infty} m(\tau) e^{-(x-\tau)^2/2\sigma_D^2} d\tau \quad (9.19)$$

where σ_D is the diffusion length. While a one-dimensional case is shown here, it is easy to extend the convolution to two or three dimensions (though things do get more complex when boundary conditions are applied).

Consider now a generic latent image for a repeating line/space pattern of pitch p described as a Fourier series:

$$m(x) = \sum_{n=0}^N a_n \cos(2\pi nx/p) \quad (9.20)$$

where a pattern symmetrical about $x = 0$ is assumed so that there are no sine terms in the series. Larger values of n represent higher-frequency terms (harmonics) in the image, though a typical minimum pitch pattern will have an upper limit of $N = 2$ or 3. The effect of diffusion on this latent image can be calculated by plugging Equation (9.20) into the convolution integral (9.19) giving

$$m^*(x) = \sum_{n=0}^N a_n^* \cos(2\pi nx/p) \quad (9.21)$$

where $a_n^* = a_n e^{-2(\pi n \sigma_D/p)^2}$.

Diffusion can be thought of as simply a reduction of each of the $n > 0$ Fourier coefficients of the latent image. Obviously, a greater diffusion length leads to a greater degradation of the latent image (a_n is reduced). But it is the diffusion length relative to the pitch that matters. Thus, for the same diffusion length, smaller pitch patterns are degraded more than larger pitch patterns. Also, the higher-frequency (larger n) terms degrade faster than the lower-frequency terms. In fact, each frequency term can be said to have an effective diffusion length equal to $n\sigma_D$ and it is the ratio of this effective diffusion length to the pitch that determines the amount of damping for that frequency component (Figure 9.11).

The effect of diffusion on the latent image gradient (LIG) is complicated by the fact that the rate of diffusion is driven by the change in the concentration gradient. Thus, in general the latent image gradient after diffusion will not be directly proportional to the gradient before diffusion – it will be influenced by the shape of the latent image to a

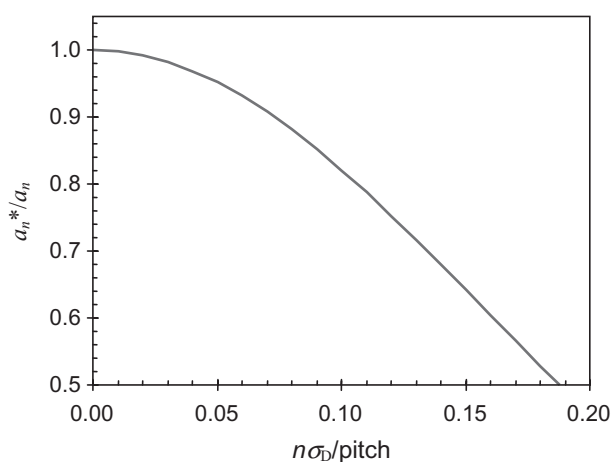


Figure 9.11 Effect of diffusion on the latent image frequency components for a dense line

distance of a few diffusion lengths away. However, an approximate response can be obtained by assuming that the latent image for a small pitch pattern, and thus the effect on the gradient, is dominated by the $n = 1$ Fourier component of the latent image. The change in the latent image gradient due to diffusion can then be approximated as the change in a_1 :

$$\frac{\partial m^*/\partial x}{\partial m/\partial x} \approx \frac{a_1^*}{a_1} \quad (9.22)$$

This simple result can be generalized for any pattern (not just patterns with a very small pitch) to become

$$\frac{\partial m^*}{\partial x} \approx \frac{\partial m}{\partial x} e^{-\sigma^2 \pi^2 / 2L^2} = \frac{\partial m}{\partial x} e^{-t_{\text{PEB}} D \pi^2 / L^2} = \frac{\partial m}{\partial x} e^{-t_{\text{PEB}} / \tau} \quad (9.23)$$

where L is a characteristic length related to the width of the edge region (the range over which the original latent image gradient is nonzero). For a pattern of small lines and spaces, L is equal to the half-pitch of the pattern. Obviously, increased diffusion (indicated by a larger diffusion length) results in a greater degradation of the latent image gradient (Figure 9.12). Also, sharper edges (smaller values of L) are more sensitive to diffusion, showing a greater fractional decline in the latent image gradient for a given diffusion length. The second (middle) form of Equation (9.23) replaces the diffusion length with $\sqrt{2Dt_{\text{PEB}}}$ where D is the diffusivity and t_{PEB} is the bake time. Finally, the left-hand form of the equation defines a characteristic pattern diffusion time $\tau = L^2/(D\pi^2)$. The bake time must be kept much less than this characteristic pattern diffusion time in order to prevent significant latent image degradation due to diffusion.

9.5.2 Chemically Amplified Resists – Reaction Only

For chemically amplified resists, diffusion during PEB is accompanied by a reaction that changes the photogenerated acid latent image into a latent image of blocked and deblocked

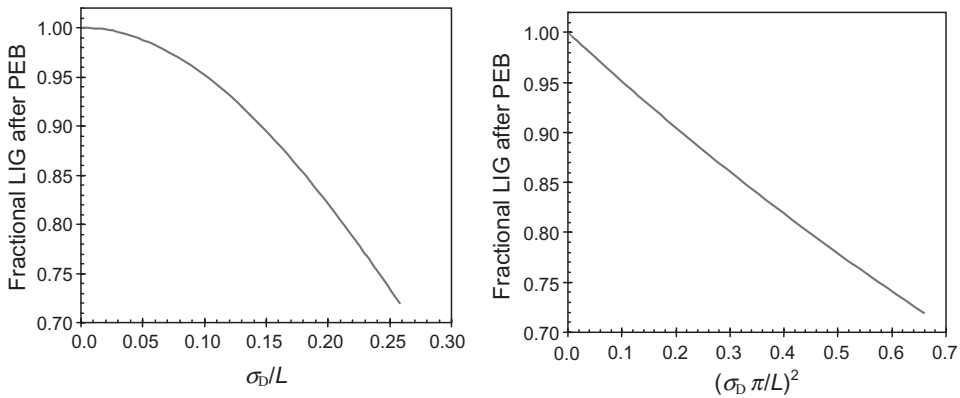


Figure 9.12 Increased diffusion (shown by the dimensionless quantity σ_D/L , the diffusion length over the width of the edge region) causes a decrease in the latent image gradient (LIG) after PEB

polymer. Since reaction and diffusion occur simultaneously, rigorous evaluation of the impact on the latent image gradient is required. However, as a start we will look at the impact of the reaction without diffusion. Ignoring the possibilities of acid loss before or during the PEB, a simple mechanism for a first-order chemical amplification would give (from Chapter 6)

$$m^*(x) = e^{-\alpha_t h(x)} \quad (9.24)$$

where $\alpha_t = K_{\text{amp}} t_{\text{PEB}}$ is the amplification factor, proportional to the PEB time and exponentially dependent on PEB temperature. This simple expression points out the trade-off between exposure dose and thermal dose. A higher exposure dose generates more acid (larger value of h), requiring less PEB (lower value of α_t) to get the same result (same value of m^*). For a given level of required amplification, thermal and exposure doses can be exchanged so long as $\alpha_t h$ is kept constant.

The gradient of this new latent image after amplification is then

$$\frac{\partial m^*}{\partial x} = m^* \ln(m^*) \left(\frac{1}{h} \frac{\partial h}{\partial x} \right) = -m^* \ln(m^*) \left(\frac{(1-h) \ln(1-h)}{h} \right) \frac{\partial \ln I}{\partial x} \quad (9.25)$$

The gradient of the deprotection latent image relative to the image log-slope is often called the chemical contrast:

$$\frac{\partial m^*}{\partial x} \bigg/ \frac{\partial \ln I}{\partial x} = \frac{\partial m^*}{\partial \ln E} = \text{chemical contrast} \quad (9.26)$$

For a given latent image after exposure [that is, given $h(x)$], the optimum latent image after amplification occurs when $m^* = e^{-1}$, giving $\alpha_t h = 1$. For a given required level of amplification (given value of m^*), the trade-off between thermal and exposure dose can be optimized to give the maximum latent image gradient after PEB. This occurs when

$$\frac{1}{h} \frac{\partial h}{\partial x} \propto \frac{(1-h) \ln(1-h)}{h} = \text{maximum when } h \rightarrow 0 \quad (9.27)$$

In other words, the optimum latent image gradient after PEB occurs when using a low dose ($h \rightarrow 0$) and a high level of amplification (Figure 9.13). Carrying this idea to its extreme, however, invalidates the assumption that no diffusion occurs, since higher levels of amplification necessitate higher levels of acid diffusion. Thus, the true trade-off between thermal and exposure dose must take into account the effects of diffusion as well.

9.5.3 Chemically Amplified Resists – Reaction–Diffusion

When diffusion is included with amplification, the situation becomes more complex. Diffusion during post-exposure bake is accompanied by a deblocking reaction that changes the solubility of the resist. Thus, it is not the final, postdiffusion distribution of exposure reaction products that controls development but rather the integral over time of these exposure products. Things become even more complicated when acid loss is accounted

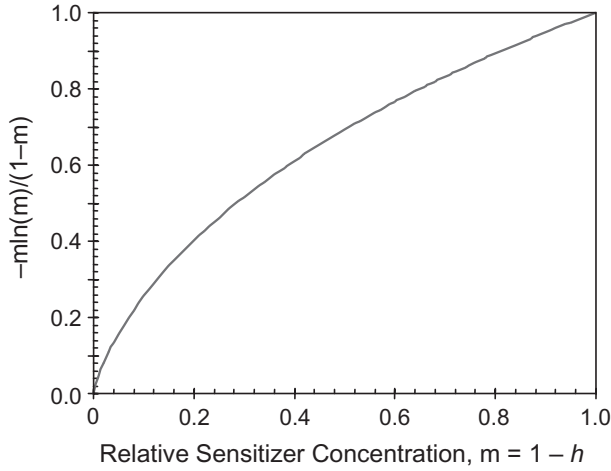


Figure 9.13 For a chemically amplified resist with a given required amount of amplification, the exposure dose (and thus relative sensitizer concentration m) is optimum as the dose approaches zero ($m \rightarrow 1$), assuming negligible diffusion

for. In particular, the presence of base quencher (that also may diffuse) leads to both complexity and advantage in tailoring the final latent image shape. However, for the simplified case of no acid loss, an analytical solution is possible.

Letting $h(x, t = 0)$ be the concentration of acid (the exposure product) at the beginning of the post-exposure bake (PEB), an effective acid latent image can be defined as

$$\begin{aligned} h_{\text{eff}}(x) &= \frac{1}{t_{\text{PEB}}} \int_0^{t_{\text{PEB}}} h(x, t = 0) \otimes \text{DPSF} dt = h(x, 0) \otimes \frac{1}{t_{\text{PEB}}} \int_0^{t_{\text{PEB}}} \text{DPSF} dt \\ &= h(x, 0) \otimes \text{RDPSF} \end{aligned} \quad (9.28)$$

where RDPSF is the reaction–diffusion point spread function (see Chapter 6). This effective acid concentration distribution $h_{\text{eff}}(x)$ can be used to calculate the reaction kinetics of the PEB as if no diffusion had taken place. In other words, the effects of diffusion are separable from the reaction for the case of no acid loss. The RDPSF then becomes analogous to the DPSF of a conventional resist.

For the 1D case, the Gaussian diffusion kernel is affected by time integration through the diffusion length, $\sigma_D = \sqrt{2Dt}$, where D is the acid diffusivity in resist. Thus,

$$\text{RDPSF} = \frac{1}{t_{\text{PEB}} \sqrt{4\pi D}} \int_0^{t_{\text{PEB}}} \frac{e^{-x^2/4Dt}}{\sqrt{t}} dt \quad (9.29)$$

The integral is solvable, resulting in an interesting final solution:

$$\text{RDPSF}(x) = 2 \frac{e^{-x^2/2\sigma_D^2}}{\sqrt{2\pi}\sigma_D} - \frac{|x|}{\sigma_D^2} \text{erfc}\left(\frac{|x|}{\sqrt{2}\sigma_D}\right) \quad (9.30)$$

The first term on the right-hand side of Equation (9.30) is nothing more than twice the *DPSF*, and thus accounts for pure diffusion. The second term, the complimentary error function times x , is a reaction term that is subtracted and thus reduces the impact of pure diffusion.

One of the most important properties of the *RDPSF* is that it falls off in x much faster than the *DPSF* with the same diffusion length. The full-width half-maximum for the *DPSF* is about $2.35\sigma_D$, but for the *RDPSF* it is about σ_D . Also, invoking a large argument approximation for the *erfc*,

$$\frac{RDPSF(x)}{DPSF(x)} \approx 2 \left(\frac{\sigma_D}{x} \right)^2, \quad x \gg \sigma_D \quad (9.31)$$

In other words, a reaction–diffusion resist system can tolerate more diffusion than a conventional resist system.

To further compare reaction–diffusion to pure diffusion, consider again the impact of diffusion (and reaction–diffusion) on the latent image. As before, we'll use a generic latent image for a repeating line/space pattern of pitch p described as a Fourier series:

$$h(x,0) = \sum_{n=0}^N a_n \cos(2\pi nx/p) \quad (9.32)$$

where a pattern symmetrical about $x = 0$ is assumed so that there are no sine terms in the series. The effect of pure diffusion, calculated as a convolution with the *DPSF*, is simply a reduction in the amplitude of each harmonic, as shown in Equation (9.21). In the reaction–diffusion case, we convolve Equation (9.32) with the *RDPSF*, which results in an analytical solution:⁵

$$h^*(x) = \sum_{n=0}^N a_n^* \cos(2\pi nx/p)$$

where

$$a_n^* = a_n \left(\frac{1 - e^{-2(\pi n \sigma_D/p)^2}}{2(\pi n \sigma_D/p)^2} \right) \quad (9.33)$$

Consider the special case of $x = p/4$ and $N = 3$ (that is, at the nominal line edge of a very small line/space pattern), a case that will prove useful later:

$$\begin{aligned} h(p/4, 0) &= a_0 - a_2 \\ h^*(p/4) &= a_0 - a_2^* \end{aligned} \quad (9.34)$$

Figure 9.14 compares the effects of pure diffusion with reaction–diffusion on the amplitudes of the Fourier coefficients. As can be seen, pure diffusion causes a much faster degradation of the Fourier components than reaction–diffusion for the same diffusion length. For example, if one is willing to allow a particular Fourier component to fall in amplitude by 20%, a reaction–diffusion system with no acid loss can tolerate about 50% more diffusion than a pure diffusion resist. In fact, for reasonably small $n\sigma_D/p$, one can

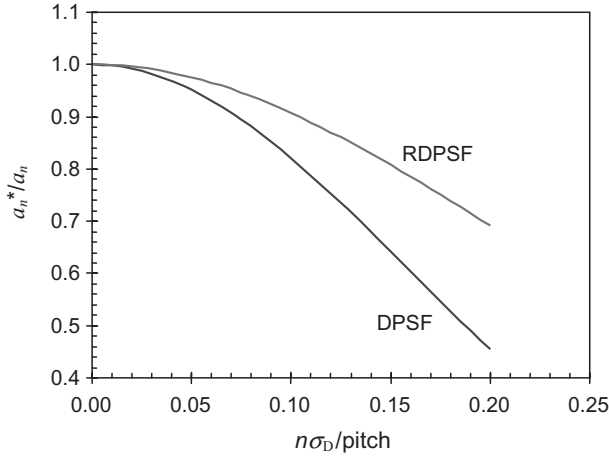


Figure 9.14 Effect of diffusion on the latent image frequency components for a dense line, comparing pure diffusion (DPSF) to reaction–diffusion (RDPSF)

compare the diffusion-only case to a reaction–diffusion system using a Taylor series expansion of the exponential:

$$\begin{aligned} \text{Diffusion: } \frac{a_n^*}{a_n} &= e^{-2(\pi n \sigma_D/p)^2} \approx 1 - 2(\pi n \sigma_D/p)^2 \\ \text{Reaction-Diffusion: } \frac{a_n^*}{a_n} &= \frac{1 - e^{-2(\pi n \sigma_D/p)^2}}{2(\pi n \sigma_D/p)^2} \approx 1 - (\pi n \sigma_D/p)^2 \end{aligned} \quad (9.35)$$

The two are roughly equivalent if the reaction–diffusion system has a diffusion length $\sqrt{2}$ larger than the diffusion-only case (or a diffusivity that is twice as large).

As with pure diffusion for conventional resist, we can approximate the impact of reaction–diffusion on the final latent image gradient by assuming the $n = 1$ order of the latent image Fourier series dominates. Thus,

$$\frac{\partial h_{\text{eff}}}{\partial x} \approx \frac{\partial h}{\partial x} \left(\frac{1 - e^{-\sigma_D^2 \pi^2 / 2L^2}}{\sigma_D^2 \pi^2 / 2L^2} \right) \quad (9.36)$$

Combining now all of the terms of exposure, reaction–diffusion impact on $h(x)$, and reaction to give m^* , the final gradient of the deblocked concentration can be calculated. The exposure term gives

$$h = 1 - e^{-CIr} \quad \text{which leads to} \quad \frac{\partial h}{\partial x} = (1 - h) \ln(1 - h) \frac{\partial \ln I}{\partial x} \quad (9.37)$$

Reaction converts the effective acid concentration to a blocked polymer concentration:

$$m^* = e^{-\alpha_f h_{\text{eff}}} \quad \text{which leads to} \quad \frac{\partial m^*}{\partial x} = -\alpha_f m^* \frac{\partial h_{\text{eff}}}{\partial x} \quad (9.38)$$

Combining Equations (9.36) through (9.38),

$$\frac{\partial m^*}{\partial x} \approx \frac{1}{\eta} m^* (1 - e^{-\eta \alpha_f}) (1 - h) \ln(1 - h) \frac{\partial \ln I}{\partial x} \quad (9.39)$$

where

$$\eta = \frac{\pi^2 \sigma_D^2}{2L^2 K_{\text{amp}} t_{\text{PEB}}} = \frac{\pi^2 D}{L^2 K_{\text{amp}}} = \frac{1}{\tau K_{\text{amp}}}$$

The term η represents the ratio of the rate of diffusion for this feature to the rate of reaction. The only remaining step is to relate h_{eff} to h at the nominal line edge. Taking the simple case described in Equation (9.34),

$$h_{\text{eff}}(p/4) = a_0 - a_2^* = a_0 - a_2 \left(\frac{1 - e^{-2(2\pi\sigma_D/p)^2}}{2(2\pi\sigma_D/p)^2} \right) \quad (9.40)$$

Expanding the exponential as a Taylor's series,

$$h_{\text{eff}} = a_0 - a_2 \left(1 - (2\pi\sigma_D/p)^2 + \frac{4}{3} (2\pi\sigma_D/p)^4 - \dots \right) \quad (9.41)$$

To avoid excessive degradation of the latent image due to diffusion, a typical value for $2\pi\sigma_D/p$ would be 0.5 or less. Typical latent images will have a_2 about one order of magnitude smaller than a_0 . Thus, one can approximate h_{eff} for the typical case as

$$h_{\text{eff}} \approx a_0 - a_2 + a_2 (2\pi\sigma_D/p)^2 \approx h \text{ to an error of } \sim 5\% \text{ or less} \quad (9.42)$$

In other words, at the line edge the acid concentration doesn't change much due to diffusion. As a result, a final expression for the latent image gradient becomes

$$\frac{\partial m^*}{\partial x} \approx \frac{1}{\eta} e^{-\alpha_f h_{\text{eff}}} (1 - e^{-\eta \alpha_f}) (1 - h_{\text{eff}}) \ln(1 - h_{\text{eff}}) \frac{\partial \ln I}{\partial x} \quad (9.43)$$

Figure 9.15 uses Equation (9.43) to show how the latent image gradient (LIG) varies with α_f (which is proportional to PEB time). At low levels of amplification (small PEB time), the gradient is amplification controlled and increased bake produces an improved gradient. At long PEB times, the gradient becomes diffusion controlled and increased bake reduces the gradient due to diffusion. Thus, there is an optimum bake time (optimum α_f) that produces the maximum LIG of blocked polymer concentration.

The goal now becomes one of finding the optimum final deblocked concentration m^* (equivalently, the optimum h_{eff}) and the optimum level of amplification α_f for a given value of η . The optimum α_f can be determined for a given value of m^* or for a given value of h_{eff} :

$$\text{optimum } \alpha_f = - \left[\frac{1 + \ln(1 - h_{\text{eff}})}{(1 - h_{\text{eff}}) \ln(1 - h_{\text{eff}})} \right] = \frac{1}{\eta} \ln \left(\frac{h_{\text{eff}} + \eta}{h_{\text{eff}}} \right) \quad (9.44)$$

Figure 9.16 provides numerical results of the values of α_f and m^* that produce the maximum latent image gradient as a function of η . Figure 9.17 shows the resulting

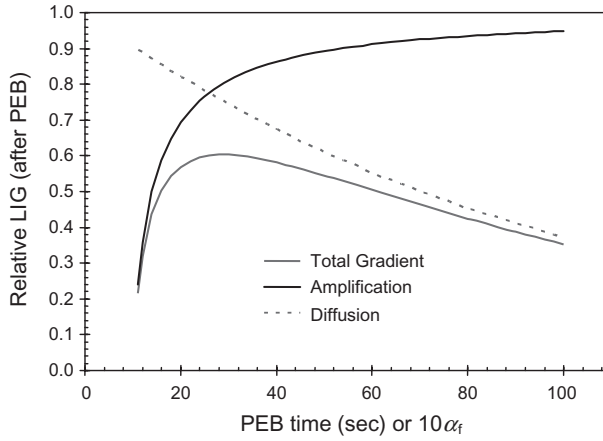


Figure 9.15 Including diffusion with amplification, there is an optimum PEB to maximize the latent image gradient (LIG), shown here relative to the maximum possible LIG. For this example, $K_{amp} = 0.1 \text{ s}^{-1}$, $\tau = 200$ seconds and the exposure dose is chosen to give the maximum gradient for each PEB time

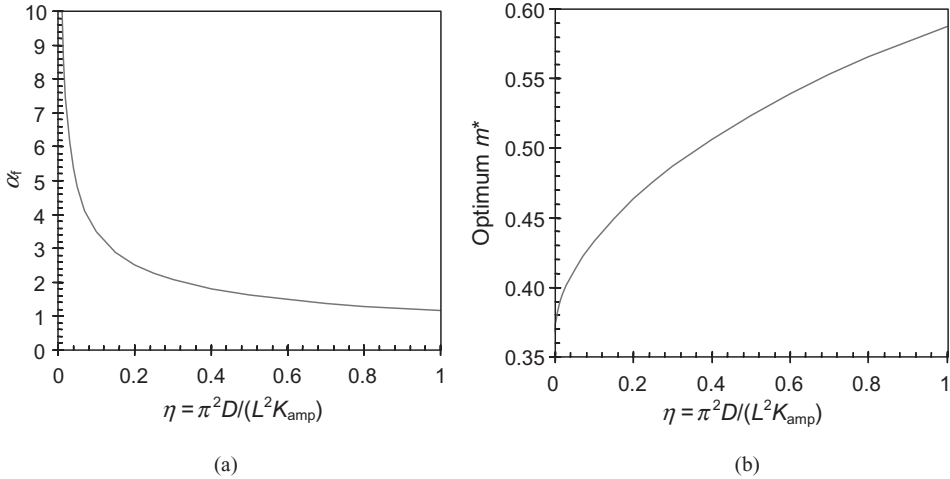


Figure 9.16 The optimum value of (a) the amplification factor, and (b) the deblocked concentration in order to maximize the final latent image gradient, as a function of η

maximum latent image gradient. For the case of $\eta = 0$ (no diffusion), the solution matches the results given in the previous section. As the relative amount of diffusion increases, the optimum amplification factor quickly falls. A knee in this response at about $\eta = 0.1$ (and $\alpha_f \approx 3$) marks a transition from high to low sensitivity of the optimum amplification factor to η . Typical values of η for commercial resists are on the order of 0.1–0.2, so that optimum amplification factors are near 3 and the optimum deblocking concentration at

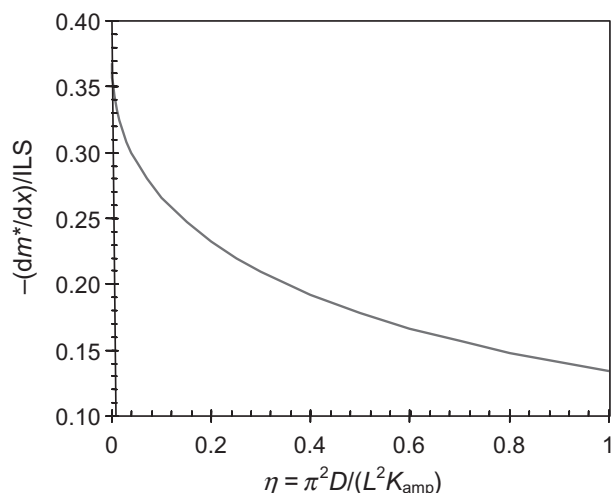


Figure 9.17 The optimum value of the final latent image gradient (relative to the image log-slope), as a function of η

the line edge is about 0.45. The optimum final LIG (relative to the image log-slope) for this range of η is about 0.25, a reduction of about 1/3 from the no-diffusion optimum LIG (see Figure 9.17).

From the results shown in Figure 9.17, it is clear that a reduction in diffusion relative to amplification (that is, a reduction in D/K_{amp} and thus η) can lead to improved final latent image gradient. Also, for a given D/K_{amp} the value of η grows as feature size shrinks so that D/K_{amp} must be continually improved with each new lithography technology generation just to keep η constant. Since both diffusivity and the amplification rate constant are temperature dependent, changing temperature will likely change their ratio. Using an Arrhenius equation for each term,

$$\frac{D}{K_{\text{amp}}} = \frac{A_{\text{rD}} e^{-E_{\text{ad}}/RT}}{A_{\text{rK}} e^{-E_{\text{ak}}/RT}} = A'_{\text{r}} e^{-(E_{\text{ad}} - E_{\text{ak}})/RT} \quad (9.45)$$

The difference in activation energies between diffusion and reaction determines whether high- or low-temperature bakes produce lower η and thus better latent image gradients.

The dependence of η with temperature is more complicated when the Byers–Petersen model is used. Recalling from Chapter 6, the overall amplification rate constant depends on the acid diffusivity as

$$K_{\text{amp}} = \frac{K_{\text{react}} K_{\text{diff}} D}{K_{\text{react}} + K_{\text{diff}} D} \quad (9.46)$$

Thus, the ratio of diffusivity to amplification becomes

$$\frac{D}{K_{\text{amp}}} = \frac{1}{K_{\text{diff}}} + \frac{D}{K_{\text{react}}} \quad (9.47)$$

In general, K_{diff} is not expected to vary much with temperature. Thus, if the resist operates in the diffusion-controlled regime ($K_{\text{react}} \gg K_{\text{diff}}D$), η will be relatively insensitive to temperature. However, in the reaction-controlled regime ($K_{\text{react}} \ll K_{\text{diff}}D$), η will have the temperature dependence shown in Equation (9.45).

9.5.4 Chemically Amplified Resists – Reaction–Diffusion with Quencher

The analysis of the previous section is very helpful for understanding the role of diffusion in a simple reaction–diffusion system with no acid loss. However, all serious chemically amplified resists include base quenchers – a beneficial source of acid loss. In general, quencher greatly complicates any efforts to analytically solve for the final latent image or the final latent image gradient. However, for one special case, the above analysis applies even when quencher is present. If the acid–quencher reaction is very fast compared to diffusion (typically a very good assumption), and the base quencher has exactly the same diffusivity as the acid, then the initial acid concentration at the start of the PEB $h(x,0)$ is simply reduced by the background quencher concentration. [Note that for a Fourier series representation of the initial acid latent image such as Equation (9.32), the impact of quencher is then just a reduction of the value of a_0 .] Negative acid concentrations represent the concentration of base quencher, so that both acid and base diffusion are properly calculated using the convolution of Equation (9.28). Of course, negative values of $h_{\text{eff}}(x)$, the effective acid concentration, should be set to zero before performing any amplification calculations.

For quencher of relative initial concentration q_0 , the acid gradient after exposure and initial quenching becomes, in regions of excess acid,

$$h = 1 - e^{-Ct} - q_0 \quad \text{which leads to} \quad \frac{\partial h}{\partial x} = (1 - h - q_0) \ln(1 - h - q_0) \frac{\partial \ln I}{\partial x} \quad (9.48)$$

The resulting final latent image gradient becomes

$$\frac{\partial m^*}{\partial x} \approx \frac{1}{\eta} e^{-\alpha_t h_{\text{eff}}} (1 - e^{-\eta \alpha_t}) (1 - h_{\text{eff}} - q_0) \ln(1 - h_{\text{eff}} - q_0) \frac{\partial \ln I}{\partial x} \quad (9.49)$$

The value of α_t to maximize the final latent image gradient is

$$\text{optimum } \alpha_t = - \left[\frac{1 + \ln(1 - h_{\text{eff}} - q_0)}{(1 - h_{\text{eff}} - q_0) \ln(1 - h_{\text{eff}} - q_0)} \right] = \frac{1}{\eta} \ln \left(\frac{h_{\text{eff}} + \eta}{h_{\text{eff}}} \right) \quad (9.50)$$

The impact of quencher on the optimum gradient is seen in Figure 9.18. Quencher allows the process to be run at higher η values while still providing a good latent image gradient. For the $q = 0.1$ case in this figure, an η value that is about 0.23 higher will give the same optimum gradient compared to the no-quencher case. Alternately, for a given η , higher q_0 allows for a higher optimum gradient. Thus, the presence of quencher significantly relaxes the constraints placed on diffusion, allowing higher-diffusing resists to work well for smaller features. Figure 9.19 shows the values of α_t and m^* that produce the optimum gradient.

Note that in Figure 9.18, the two curves for nonzero quencher are only plotted down to some critical value of η (called η_c). When $\eta < \eta_c$, the optimum m^* approaches 1 (see

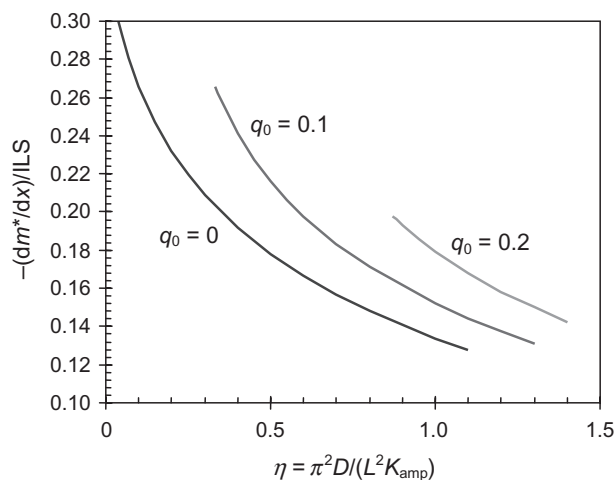


Figure 9.18 The optimum value of the final latent image gradient (relative to the image log-slope), as a function of η for cases with and without quencher

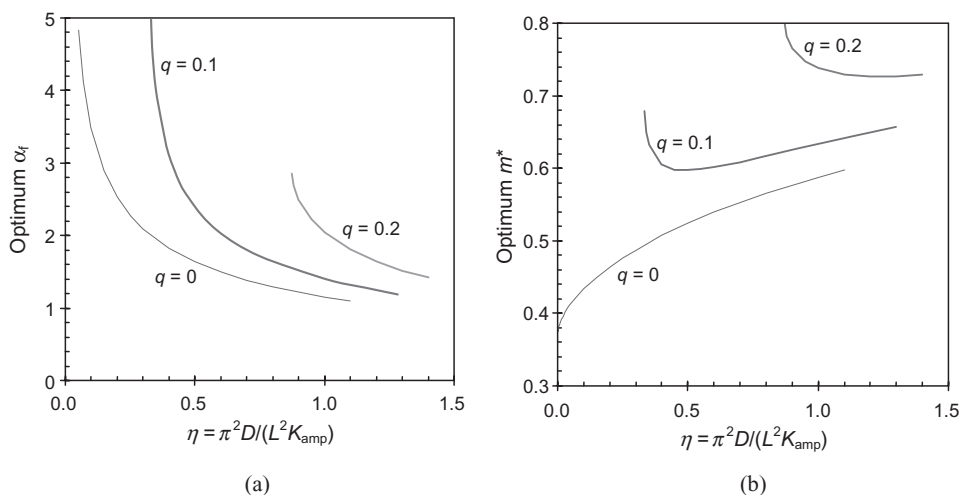


Figure 9.19 The optimum value of (a) the amplification factor, and (b) the deblocked concentration in order to maximize the final latent image gradient, as a function of η for different quencher loadings

Figure 9.19). For this case, the best latent image gradient occurs when the point at which quencher just eliminates all acid (that is, $h = 0$) is placed at the nominal line edge. The resulting LIG is quite high (see Problem 9.8):

$$\left. \frac{\partial m^*/\partial x}{ILS} \right|_{\max} \approx \frac{1}{\eta} (1 - q_0) \ln(1 - q_0) \quad (9.51)$$

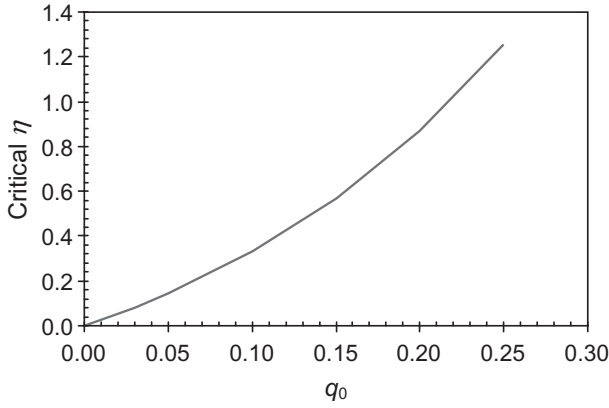


Figure 9.20 Numerically determined values of the critical η value (η_c) for different quencher loadings

Of course, real development processes require some level of deblocking at the line edge so that this maximum gradient cannot be practically attained. Nonetheless, operating below η_c and putting the $h = 0$ point very near the nominal line edge result in very steep latent image gradients. A plot of η_c as a function of quencher loading is shown in Figure 9.20.

9.6 Develop

In one sense, the goal of selective exposure of resist with an aerial image is to create a solubility differential: exposed and unexposed regions of the photoresist give rise to regions of higher and lower solubility, measured as a development rate (resist removal rate in nanometers per second). The information contained in the aerial image (or image in resist) $I(x)$ is used to expose the photoresist to form a latent image $m(x)$ or $h(x)$, which is modified by the post-exposure bake to create a new latent image $m^*(x)$, and finally developed based on a development rate ‘image’ $r(x)$ that results in the definition of the feature edge.

The fundamental chemical response of interest is the change in dissolution rate as a function of the exposure dose seen by the resist. A plot of development rate r versus exposure dose E on a log-log scale is called a Hurter–Driffield (H–D) curve (Figure 9.21) and allows for the definition of the *photoresist contrast*, γ . Quite simply, the photoresist contrast is the slope of the development rate H–D curve (see Chapter 7).

$$\gamma \equiv \frac{\partial \ln r}{\partial \ln E} \quad (9.52)$$

Note that photoresist contrast is actually a function of exposure, $\gamma(E)$.

The photoresist contrast is a measure of the discrimination of the resist with respect to exposure. Higher contrast means that a given change in dose will result in a greater change

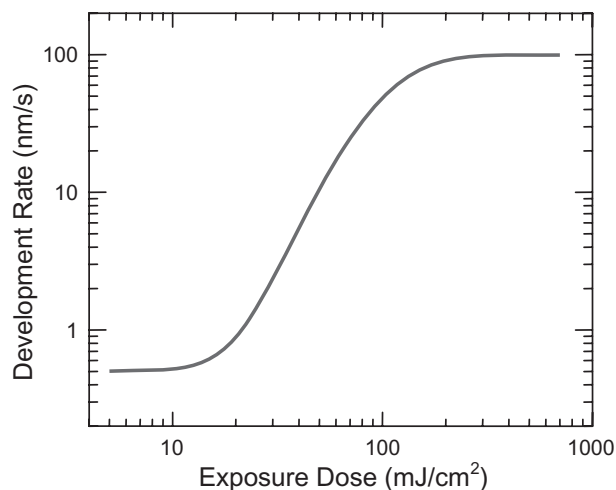


Figure 9.21 Typical development rate function of a positive photoresist (one type of Hurter–Driffield curve)

in development rate. This point can be seen clearly using the *lithographic imaging equation*, derived from the definition of photoresist contrast:

$$\frac{\partial \ln r}{\partial x} = \gamma(E) \frac{\partial \ln I}{\partial x} \quad \text{or} \quad w \frac{\partial \ln r}{\partial x} = \gamma N I L S \quad (9.53)$$

The quality of the development rate image, described by a gradient in the dissolution rate, is determined by the product of the image log-slope and the photoresist contrast function. To create a large solubility differential, one would like a good aerial image (large image log-slope), a good photoresist (large photoresist contrast) and an optimized process (an exposure dose chosen to use the maximum of the photoresist contrast function).

Breaking up the response of development to exposure into separate latent and development rate images allows us to relate the development rate gradient to the latent image gradient:

$$\frac{\partial \ln r}{\partial x} = \frac{\partial \ln r}{\partial m^*} \frac{\partial m^*}{\partial x} \quad (9.54)$$

where m^* is the concentration of the chemical species (after post-exposure bake) that affects the dissolution rate. For example, for a chemically amplified resist m^* would represent the concentration of blocked polymer. From Equations (9.53) and (9.54), one can see that the definition of photoresist contrast encompasses the exposure, post-exposure bake and development steps in order to relate the final development rate gradient to the original source of the information being imprinted in the resist, the image log-slope.

The variation of development rate with m^* can take on many forms, but a simple one can be used here to illustrate the expected response. The model of development shown

here (called the reaction-controlled version of the original kinetic development model) provides for a typical nonlinear development rate function:

$$r = r_{\max}(1 - m^*)^n + r_{\min} \quad (9.55)$$

where $r_{\max}$ and $r_{\min}$ represent the maximum and minimum development rates, respectively ($r_{\max} \gg r_{\min}$ is assumed), and n is called the dissolution selectivity parameter and controls how nonlinear the development response will be. Using this model of dissolution rate,

$$\frac{\partial \ln r}{\partial m^*} = -\frac{n}{1 - m^*} \left(1 - \frac{r_{\min}}{r} \right) \quad (9.56)$$

Figure 9.22 shows a plot of Equation (9.56) as a function of m^* for $n = 5$ (a low to medium contrast resist) and $n = 10$ (a medium to high contrast resist). Adjusting the process to set the peak of this curve at the nominal feature edge is a key part of process optimization.

From Equation (9.56), it is easy to show that the maximum gradient occurs when

$$-\frac{\partial \ln r}{\partial m^*} \text{ is a maximum when } r = nr_{\min} \quad (9.57)$$

The maximum gradient will occur at a value of m^* given by

$$-\frac{\partial \ln r}{\partial m^*} \Big|_{\max} \text{ when } m^* = 1 - \left[\frac{(n-1)r_{\min}}{r_{\max}} \right]^{\frac{1}{n}} \quad (9.58)$$

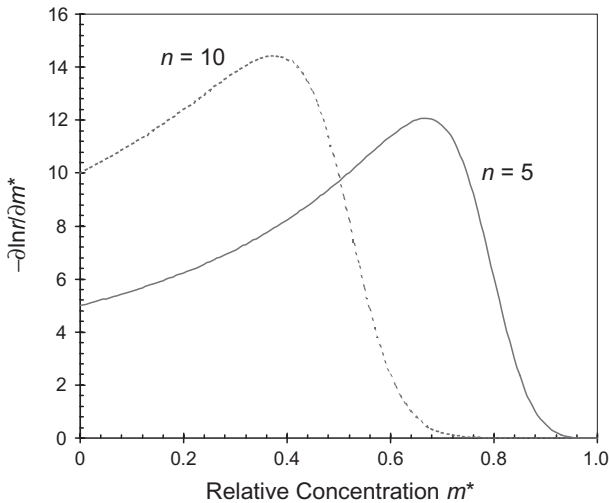


Figure 9.22 One component of the overall photoresist contrast is the variation in development rate r with chemical species m^* , shown here for the reaction-controlled version of the original kinetic development rate model ($r_{\max} = 100 \text{ nm/s}$, $r_{\min} = 0.1 \text{ nm/s}$)

Thus, the maximum gradient is

$$-\frac{\partial \ln r}{\partial m^*} \Big|_{\max} = \frac{(n-1)}{1-m^*} = (n-1) \left[\frac{r_{\max}}{(n-1)r_{\min}} \right]^{\frac{1}{n}} \quad (9.59)$$

For the full kinetic rate model,

$$r = r_{\max} \frac{(a+1)(1-m^*)^n}{a + (1-m^*)^n} + r_{\min} \quad (9.60)$$

and the equivalent gradient is

$$\frac{\partial \ln r}{\partial m^*} = -\frac{n}{1-m^*} \left(1 - \frac{r_{\min}}{r} \right) \left(\frac{a}{a + (1-m^*)^n} \right) \quad (9.61)$$

Figure 9.23 shows that again there will be an optimum inhibitor concentration to maximize the development rate gradient, and that this optimum will depend on the value of the dissolution selectivity parameter. The maximum gradient occurs when the development rate is given by

$$r = \frac{nr_{\min}}{1 + (n-1) \frac{(1-m^*)^n}{(1-m_{\text{th}})^n}} \quad (9.62)$$

The maximum gradient is then

$$-\frac{\partial \ln r}{\partial m^*} \Big|_{\max} = \frac{(n-1)}{(1-m^*)} \left(\frac{1 - \frac{(1-m^*)^n}{(1-m_{\text{th}})^n}}{1 + \left(\frac{n-1}{n+1} \right) \frac{(1-m^*)^n}{(1-m_{\text{th}})^n}} \right) \quad (9.63)$$

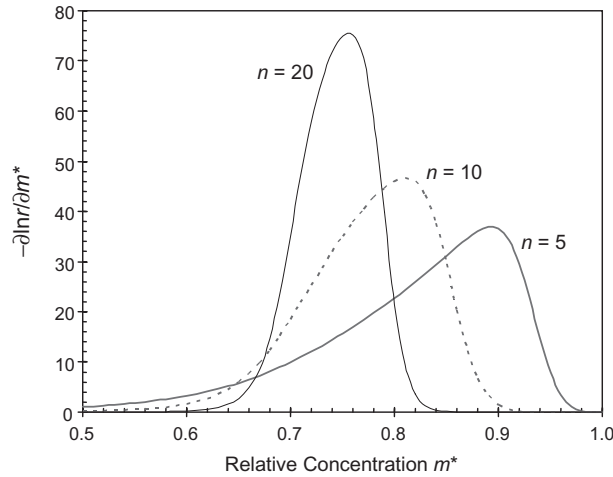


Figure 9.23 Development rate gradient with inhibitor concentration for the original kinetic development rate model ($r_{\max} = 100 \text{ nm/s}$, $r_{\min} = 0.1 \text{ nm/s}$, $m_{\text{th}} = 0.7$)

As n increases, the optimum inhibitor concentration approaches the threshold inhibitor concentration, m_{th} . For large n , the maximum gradient occurs when

$$1 - m^* \approx (1 - m_{\text{th}}) \left(1 + \frac{1}{2n} \ln \left(\frac{r_{\text{min}}}{r_{\text{max}}} \right) \right) \quad \text{and} \quad \frac{(1 - m^*)^n}{(1 - m_{\text{th}})^n} \approx \sqrt{\frac{r_{\text{min}}}{r_{\text{max}}}} \quad (9.64)$$

giving

$$-\left. \frac{\partial \ln r}{\partial m^*} \right|_{\text{max}} \approx \frac{(n-1)}{(1 - m_{\text{th}}) \left(1 + \frac{1}{2n} \ln \left(\frac{r_{\text{min}}}{r_{\text{max}}} \right) \right)} \left(\frac{1 - \sqrt{\frac{r_{\text{min}}}{r_{\text{max}}}}}{1 + \left(\frac{n-1}{n+1} \right) \sqrt{\frac{r_{\text{min}}}{r_{\text{max}}}}} \right) \quad (9.65)$$

If the resist is a reasonably good one, so that $r_{\text{max}}/r_{\text{min}} > 1000$, then this large n approximation can be further simplified:

$$-\left. \frac{\partial \ln r}{\partial m^*} \right|_{\text{max}} \approx \frac{n}{(1 - m_{\text{th}})} + \frac{1}{2(1 - m_{\text{th}})} \ln \left(\frac{r_{\text{max}}}{r_{\text{min}}} \right) \quad (9.66)$$

This final result describes the basic behavior that we wish to know: the maximum log-development rate gradient is roughly proportional to n , inversely proportional to $1 - m_{\text{th}}$, and increases with the log of the ratio of the maximum to minimum development rates.

9.6.1 Conventional Resist

While Equation (9.54) related the development rate gradient to the after-PEB latent image gradient, in fact all of the lithographic process steps can be chained together to relate the final development response to the initial source of the imaging information, the aerial image. For a conventional resist,

$$\gamma = \left(\frac{\partial \ln r}{\partial m^*} \right) \left(\frac{\partial m^*}{\partial m} \right) \left(\frac{\partial m}{\partial \ln E} \right) \quad (9.67)$$

As described above, we have seen that for a resist that does not bleach,

$$\frac{\partial m}{\partial \ln E} = m \ln(m) \quad (9.68)$$

The impact of diffusion can approximated by

$$\frac{\partial m^*}{\partial m} \approx e^{-\sigma^2 \pi^2 / 2L^2} \quad (9.69)$$

where σ is the diffusion length and L is a characteristic width of the edge region (and is roughly equal to the resolution of the lithography process). Also, diffusion does not have a significant impact on the concentration of the sensitizer at the line edge so that $m \approx m^*$

at the feature edge. Combining each of these expressions with Equation (9.61) gives the overall contrast:

$$\gamma = -\frac{n}{1-m} \left(1 - \frac{r_{\min}}{r}\right) \left(\frac{a}{a + (1-m)^n}\right) (e^{-\sigma^2 \pi^2 / 2L^2}) m \ln m \quad (9.70)$$

Our goal will now be to find the maximum value of this photoresist contrast. Since diffusion always degrades the contrast, the optimum will always be zero diffusivity. Thus, without loss of generality, we shall omit the diffusion term from the analysis that follows. The exact value of the maximum is most easily determined numerically (Figure 9.24), but it will be useful to derive some general trends. Since the $m \ln m$ term varies slowly, the contrast will be dominated by the development piece ($\partial \ln r / \partial m^*$). Thus, for reasonably large n the optimum value of m^* will be approximately that given by Equation (9.64).

$$m^* \approx m_{\text{th}} + \frac{(1-m_{\text{th}})}{n} \ln \left(\sqrt{\frac{r_{\max}}{r_{\min}}} \right) \quad (9.71)$$

Thus, at this optimum

$$m^* \ln m^* \approx m_{\text{th}} \ln m_{\text{th}} + \frac{(1 + \ln m_{\text{th}})(1-m_{\text{th}})}{n} \ln \left(\sqrt{\frac{r_{\max}}{r_{\min}}} \right) \quad (9.72)$$

The maximum contrast will be approximately

$$\gamma_{\max} \approx -\frac{m_{\text{th}} \ln m_{\text{th}}}{(1-m_{\text{th}})} \left[n + \left(\frac{(1-m_{\text{th}} + \ln m_{\text{th}})}{m_{\text{th}} \ln m_{\text{th}}} \right) \ln \left(\sqrt{\frac{r_{\max}}{r_{\min}}} \right) \right] (e^{-\sigma^2 \pi^2 / 2L^2}) \quad (9.73)$$

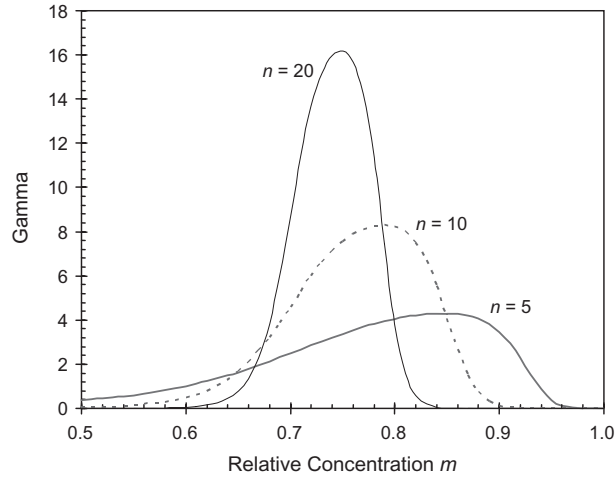


Figure 9.24 Photoresist contrast as a function of inhibitor concentration for the original kinetic development rate model, assuming no diffusion ($r_{\max} = 100 \text{ nm/s}$, $r_{\min} = 0.1 \text{ nm/s}$, $m_{\text{th}} = 0.7$) and different values of the dissolution selectivity parameter n

or, since n is reasonably high,

$$\gamma_{\max} \approx -n \frac{m_{\text{th}} \ln m_{\text{th}}}{(1 - m_{\text{th}})} (e^{-\sigma^2 \pi^2 / 2L^2}) \quad (9.74)$$

This final approximate form matches well the full numerical calculations for the maximum contrast shown in Figure 9.24 even for n as low as 5. The overall contrast is maximized by minimizing diffusion, maximizing n and letting m_{th} approach 1.

9.6.2 Chemically Amplified Resist

For a chemically amplified resist, the PEB term $\partial m^* / \partial m$ is a function of the amplification factor α_t and the diffusion factor η , as described previously. Figure 9.25 shows a plot of $\gamma(E, \alpha_t)$ for the $n = 5$ case assuming no diffusion and the simple reaction-controlled development rate equation. Without diffusion, the best performance comes from a low dose and a high amplification factor, as described previously.

When diffusion is included in the analysis, the best resist will be the one that matches the optimum blocked concentration to produce a good latent image gradient (for example, see the curves in Figure 9.19b) with the optimum blocked concentration to maximize $\partial \ln r / \partial m^*$ (for example, see the curves in Figure 9.23). Thus, the m_{th} of the development function must be tuned to the value of η and quencher concentration for the resist. And,

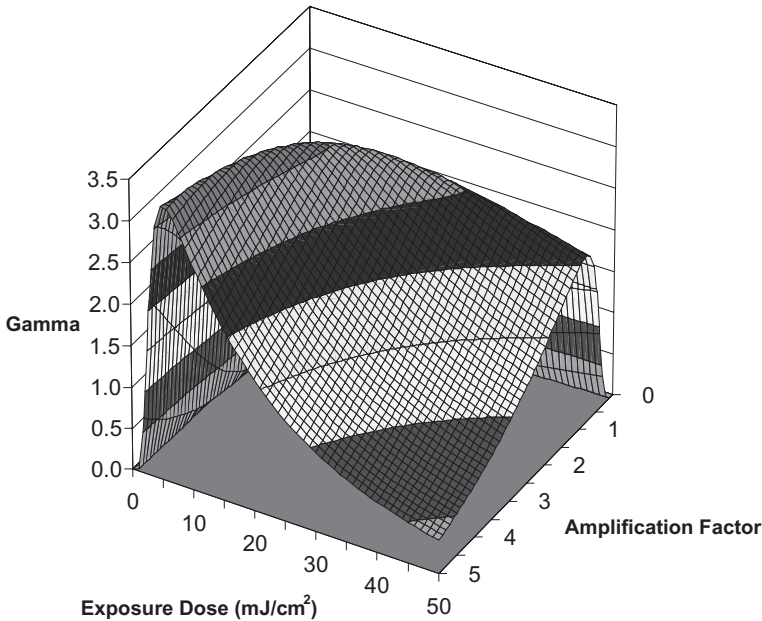


Figure 9.25 The overall photoresist contrast (gamma) as a function of exposure dose (E) and amplification factor (α_t) for a chemically amplified resist with no diffusion ($\eta = 0$, $r_{\max} = 100 \text{ nm/s}$, $r_{\min} = 0.1 \text{ nm/s}$, $n = 5$, $C = 0.05 \text{ cm}^2/\text{mJ}$)

if the quencher level is set high enough so that η is near its critical level, the maximum resist contrast can approach its maximum possible level:

$$\gamma_{\max} \approx n \frac{m_{\text{th}}}{(1 - m_{\text{th}})} \left[\frac{(1 - q_0) \ln(1 - q_0)}{\eta} \right], \quad m_{\text{th}} \approx e^{-\alpha_t h_{\text{eff}}}. \quad (9.75)$$

9.7 Resist Profile Formation

We have now arrived at the final ‘image’ gradient, the gradient in development rate. As we shall see next, the last step will be to relate this gradient to the control of the resist feature edge by knowing the development path. The development path traces the surface of the resist through the development cycle from the top of the undeveloped resist to a point on the final resist profile (see Chapter 7). The basic equation which defines the physical process of development is an integral equation of motion:

$$t_{\text{dev}} = \oint \frac{ds}{r(x, y, z)} \quad (9.76)$$

where t_{dev} is the development time, $r(x, y, z)$ is the development rate at every point in the resist, ds is a differential length along the development path, and the path integral is taken along the development path. The endpoint of the development path defines the position of the final resist profile and, consequently, the final critical dimension. There is only one path possible, determined by the principle of least action: the path will be that which goes from start to end in the least amount of time.

In Chapter 7, several solutions for the development path were given for certain special cases. Two of these cases are described below.

9.7.1 The Case of a Separable Development Rate Function

An interesting and useful case to examine is when the development rate function $r(x, z)$ is separable into a function of incident dose E only and a function of the spatial variables independent of dose:

$$r(E, x, z) = f(E)g(x, z) \quad (9.77)$$

In general, none of the kinetic development models discussed in Chapter 7 meet this criterion. However, separability results from assuming a constant theoretical contrast. Integrating the definition of theoretical contrast for the case of constant contrast gives

$$r(x, z) \propto E^\gamma (I_r(x, z))^\gamma \quad (9.78)$$

where $I_r(x, z)$ is the image in resist. Thus, separability can be assumed if the range of dose seen over the development path is sufficiently small to fall within the approximately linear region of the H–D curve.

As we saw in Chapter 7, separability leads to the interesting result that the path of development does not change with dose – the development just follows this path at a different rate. Applying this idea to the defining development path Equation (9.76),

$$f(E)t_{\text{dev}} = \int_{x_0}^x \frac{dx \sqrt{1 + z'^2}}{g(x, z)} \quad (9.79)$$

For example, let $f(E) = E^\gamma$. If the dose is increased, the same path is used to get from x_0 to x , but it arrives in a shorter time.

Equation (9.79) allows us to calculate exposure latitude: how a change in dose effects the final position on the path x for a given development time. Taking the derivative of this equation with respect to log-exposure dose,

$$\frac{\partial f}{\partial \ln E} t_{\text{dev}} = \frac{\sqrt{1+z'^2}}{g(x,z)} \frac{\partial x}{\partial \ln E} \quad (9.80)$$

Solving for the exposure latitude term, $\partial x / \partial \ln E$,

$$\frac{\partial x}{\partial \ln E} = \frac{\partial \ln f}{\partial \ln E} \frac{r(x,z)t_{\text{dev}}}{\sqrt{1+z'^2}} = \frac{\partial \ln f}{\partial \ln E} r(x,z)t_{\text{dev}} \sin \theta \quad (9.81)$$

where θ is the sidewall angle of the resist profile (which generally is close to 90°). For the case of $f(E) = E^\gamma$,

$$\frac{\partial x}{\partial \ln E} = \gamma r(x,z)t_{\text{dev}} \sin \theta \quad (9.82)$$

Similarly, the sensitivity of the final edge position x to changes in development time can also be calculated. Taking the derivative of Equation (9.79) with respect to log-development time,

$$\frac{\partial x}{\partial \ln t_{\text{dev}}} = r(x,z)t_{\text{dev}} \sin \theta \quad (9.83)$$

While Equations (9.82) and (9.83) were derived to give us insight into exposure latitude and development latitude, they also provide for an interesting approach to understanding the development time/exposure dose trade-off. If a longer development time is used, a lower dose can be used. But what combination of dose and development time will maximize CD control? For our separable development rate assumption, Equation (9.79) shows us that any process that gives the correct edge position (correct CD) must have $r(x,z)t_{\text{dev}} = \text{constant}$. But Equations (9.82) and (9.83) show us that $r(x,z)t_{\text{dev}} = \text{constant}$ will produce a constant sensitivity to relative dose errors and relative development time errors. In other words, under the assumption of constant γ , any dose/develop time combination that produces the correct CD will also allow for the control of that CD with the same sensitivity to relative dose errors and relative development time errors.

9.7.2 Lumped Parameter Model

In Chapter 7, the lumped parameter model (LPM) was used to derive a general way to relate the dose $E(x)$ required to achieve a certain CD ($=2x$) to the shape of the aerial image $I(x)$, using resist contrast γ and the effective resist thickness D_{eff} to describe the influence of the photoresist on this image:

$$\left(\frac{E(x)}{E(0)} \right)^\gamma = 1 + \frac{1}{D_{\text{eff}}} \int_{x_0}^x \left(\frac{I(x')}{I(x_0)} \right)^{-\gamma} dx' \quad (9.84)$$

where the starting point in the development path, x_0 , is generally assumed to be the middle of the space for a pattern with narrow spaces. From this, it is easy to differentiate and calculate the exposure latitude of the resist edge position x :

$$\frac{\partial x}{\partial \ln E} = r(x)\gamma t_{\text{dev}} = \gamma D_{\text{eff}} \left[\frac{E(x)I(x)}{E(0)I(0)} \right]^\gamma \quad (9.85)$$

Note that the result matches the general expression (9.82) since the LPM makes the explicit assumption of segmented development so that the development path will always be horizontal and $\sin \theta = 1$.

Equation (9.85) can be used for a specific case also derived in Chapter 7. Assuming that the shape of the aerial image in the region being developed can be approximated as a Gaussian (that is, assuming the log-slope of the image varies linearly with position in this region), the LPM predicts

$$\left(\frac{E(x)}{E(0)} \right)^\gamma \approx 1 + \left(\frac{1}{\gamma D_{\text{eff}} ILS} \right) \left(\frac{I(x)}{I(0)} \right)^{-\gamma} \quad (9.86)$$

where the approximate equal sign is used since the actual equation yields a Dawson's integral that is approximated in Equation (9.86). Note that the image log-slope, ILS , is used in this equation as a function of x rather than being defined at the nominal line edge. Substituting this specific result into the general LPM expression for exposure latitude gives

$$\frac{\partial x}{\partial \ln E} \approx \frac{1}{ILS} + \gamma D_{\text{eff}} \left[\frac{I(x)}{I(0)} \right]^\gamma \quad (9.87)$$

Often it is convenient to substitute $CD = 2x$ and compute the log-log slope of the CD-versus-exposure dose curve:

$$\frac{\partial \ln CD}{\partial \ln E} \approx \frac{2}{NILS} + \frac{2\gamma D_{\text{eff}}}{CD} \left[\frac{I(CD/2)}{I(0)} \right]^\gamma \quad (9.88)$$

There are two distinct terms on the right-hand side of this expression. The first, $2/NILS$, is a pure aerial image term and is the limiting value of the exposure latitude for the case of an infinite contrast resist (Figure 9.26). The second term is a 'development path factor' that includes the aspect ratio of the resist (D_{eff}/CD) and the ratio of the aerial image intensity at the edge of the pattern relative to that in the center of the space (that is, the exposure dose at the end of the path relative to the exposure dose at the beginning of the path). This development path factor is reduced (giving better CD control) by lowering the aspect ratio of the resist, increasing the resist contrast, and reducing the aerial image intensity at the dark line edge (i.e. $x = CD/2$) relative to the bright space center ($x = 0$).

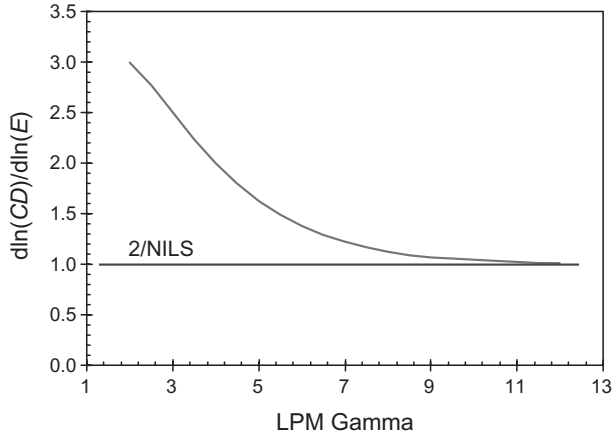


Figure 9.26 A plot of Equation (9.88) showing how the exposure latitude term approaches its limiting value of $2/NILS$ as the lumped photoresist contrast increases. In this case, the resist aspect ratio is 2, the ratio $I(CD/2)/I(0)$ is 0.5 and the NILS is 2

A slightly more accurate form of Equation (9.88) can be obtained by using two terms in the polynomial expansion of the Dawson's integral. The result is

$$\frac{\partial \ln CD}{\partial \ln E} \approx \frac{2}{NILS} \left(1 + \frac{2}{\gamma NILS} + \gamma NILS \frac{D_{\text{eff}}}{CD} \left[\frac{I(CD/2)}{I(0)} \right]^\gamma \right) \quad (9.89)$$

The relative difference between this equation and Equation (9.88) is generally very small except for the case where the resist is very thin.

Equation (9.88) finally brings all the pieces of the NILS puzzle together, describing the information transfer from the aerial image through development to the final resist image. It relates the fractional change in CD to the fractional change in exposure dose and thus its inverse defines the exposure latitude. The aerial image affects CD control in two ways, through the NILS directly and through the development path factor. The exposure, PEB and development effects can be lumped together into a photoresist contrast term, or can be separated out into individual components as described earlier.

Since the above expressions were derived specifically for a Gaussian aerial image, that specific image can be used in Equation (9.88) or (9.89).

$$\frac{I(CD/2)}{I(0)} = e^{-NILS/4} \quad (9.90)$$

giving

$$\frac{\partial \ln CD}{\partial \ln E} \approx \frac{2}{NILS} \left(1 + \frac{2}{\gamma NILS} + \gamma NILS \frac{D_{\text{eff}}}{CD} e^{-\gamma NILS/4} \right) \quad (9.91)$$

In this form, the exposure latitude is determined by just three factors: NILS, resist contrast and the feature aspect ratio. For large $\gamma NILS$, Equation (9.91) approaches the result for

an ideal threshold resist. Also, recalling the lithographic imaging equation, γ_{NILS} is the normalized log-development rate gradient. Thus, optimizing this gradient helps make the photoresist response approach that of an ideal resist.

9.8 Line Edge Roughness

When variations in the width of a resist feature occur quickly over the length of the feature, this variation is called *line width roughness* (see Figure 9.27). When examining these variations along just one edge, it is called *line edge roughness* (LER). LER becomes important for feature sizes on the order of 100 nm or less, and can become a significant source of linewidth control problems for features below 50 nm. LER is caused by a number of statistically fluctuating effects at these small dimensions such as shot noise (photon flux variations), statistical distributions of chemical species in the resist such as photoacid generators (PAGs), the random walk nature of acid diffusion during chemical amplification, and the nonzero size of resist polymers being dissolved during development.

LER is usually characterized as the 3σ deviation of a line edge from a straight line, though a more complete frequency analysis of the roughness can be valuable as well. For 193-nm lithography, LER values of 4 nm and larger are common. The impact of LER on device performance depends on the specific device layer and specific aspects of the device technology. For lithography generations below 100 nm, typical specifications for the 3σ LER are about 5% of the nominal CD. It is possible that LER will become the main limiter of CD control below 65-nm production.

In Chapter 6, a stochastic model for exposure and reaction–diffusion of chemically amplified resists was developed. This stochastic model will now prove useful for the prediction of line edge roughness. As a review, for a given volume of resist under consideration, the statistical variance in the final blocked polymer concentration is given by

$$\sigma_{m^*}^2 = \frac{\langle m^* \rangle}{\langle n_{0-\text{blocked}} \rangle} + (\langle m^* \rangle \ln \langle m^* \rangle)^2 \left(\frac{a}{\sigma_D} \right)^p \left(\frac{1}{\langle n_{0-\text{PAG}} \rangle \langle h \rangle} + \frac{[(1 - \langle h \rangle) \ln(1 - \langle h \rangle)]^2}{\langle h \rangle^2 \langle n \rangle} \right) \quad (9.92)$$

where $\langle h \rangle$ is the mean value of the acid concentration, $\langle m^* \rangle$ is the mean value of the blocked polymer concentration after PEB, σ_D/a is the ratio of acid diffusion length to the

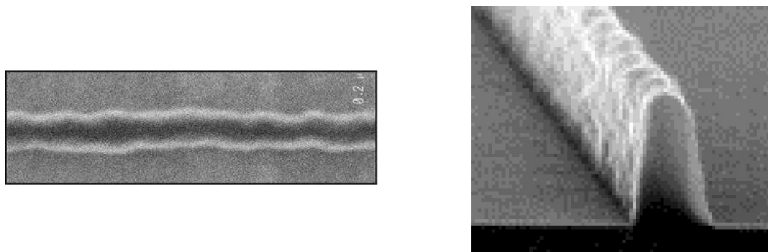


Figure 9.27 SEM pictures of photoresist features exhibiting line edge roughness

capture range of the deblocking reaction, $\langle n \rangle$ is the mean number of photons (to account for photon shot noise), $\langle n_{0-\text{PAG}} \rangle$ is the mean initial number of PAGs in the control volume at the start of exposure, and $\langle n_{0-\text{blocked}} \rangle$ is the mean initial number of blocked polymer groups in the volume before PEB.

While development can also be included, for the sake of simplicity, we will assume an infinite contrast development process so that the line edge will be determined by the blocked polymer latent image. Thus, a simple threshold model for the latent image will determine the resist critical dimension. And, as we saw in Chapter 2 for an aerial image threshold model, a Taylor series expansion of the blocked polymer concentration, cut off after the linear term, allows us to predict how a small change in blocked polymer concentration (Δm^*) will result in a change in edge position (Δx):

$$\Delta x = \frac{\Delta m^*}{dm^*/dx} \quad (9.93)$$

From this, we can devise a simple qualitative model for line edge roughness. The standard measure of line edge roughness, from a top-down SEM, will be proportional to the standard deviation of blocked polymer concentration divided by its gradient perpendicular to the line edge:

$$\text{LER} \propto \frac{\sigma_{m^*}}{dm^*/dx} \quad (9.94)$$

Thus, to achieve a low LER it will be necessary to make the standard deviation of the deprotection small and make the gradient of deprotection large. Of course, a main topic of this chapter was how process parameters can be used to maximize the latent image gradient. From Equation (9.92) we can see how to minimize the statistical uncertainty in deprotection. There is one interesting variable in common to both: acid diffusion. Increasing acid diffusion will reduce σ_{m^*} , but will reduce the latent image gradient. One would expect, then, an optimum level of diffusion to minimize the LER.

To investigate the impact of diffusion on LER, we can combine Equations (9.92) and (9.43) into (9.94). Thus, for the no-quencher case, and ignoring photon shot noise,

$$\begin{aligned} \sigma_{m^*} &= \sqrt{\frac{\langle m^* \rangle}{\langle n_{0-\text{blocked}} \rangle} + (\langle m^* \rangle \ln \langle m^* \rangle)^2 \left(\frac{a}{\sigma_D} \right)^p \left(\frac{1}{\langle n_{0-\text{PAG}} \rangle \langle h \rangle} \right)} \\ \frac{\partial m^*}{\partial x} &\propto \frac{1}{\sigma_D^2} (1 - e^{-\pi^2 \sigma_D^2 / 2L^2}) \approx \frac{\pi^2}{2L^2} \left(1 - \frac{\pi^2 \sigma_D^2}{4L^2} \right) \text{ (since } \sigma_D \ll L \text{)} \end{aligned} \quad (9.95)$$

so that

$$\text{LER} \propto \frac{\sigma_D^2}{1 - e^{-\pi^2 \sigma_D^2 / 2L^2}} \sqrt{\frac{\langle m^* \rangle}{\langle n_{0-\text{blocked}} \rangle} + (\langle m^* \rangle \ln \langle m^* \rangle)^2 \left(\frac{a}{\sigma_D} \right)^p \left(\frac{1}{\langle n_{0-\text{PAG}} \rangle \langle h \rangle} \right)} \quad (9.96)$$

and

$$\text{LER} \propto \frac{\sigma_D^2}{1 - e^{-\pi^2 \sigma_D^2 / 2L^2}} \sqrt{1 - (K_{\text{amp}} t_{\text{PEB}}) \langle m^* \rangle \ln \langle m^* \rangle \left(\frac{a}{\sigma_D} \right)^p \frac{\langle n_{0-\text{block}} \rangle}{\langle n_{0-\text{PAG}} \rangle}} \quad (9.97)$$

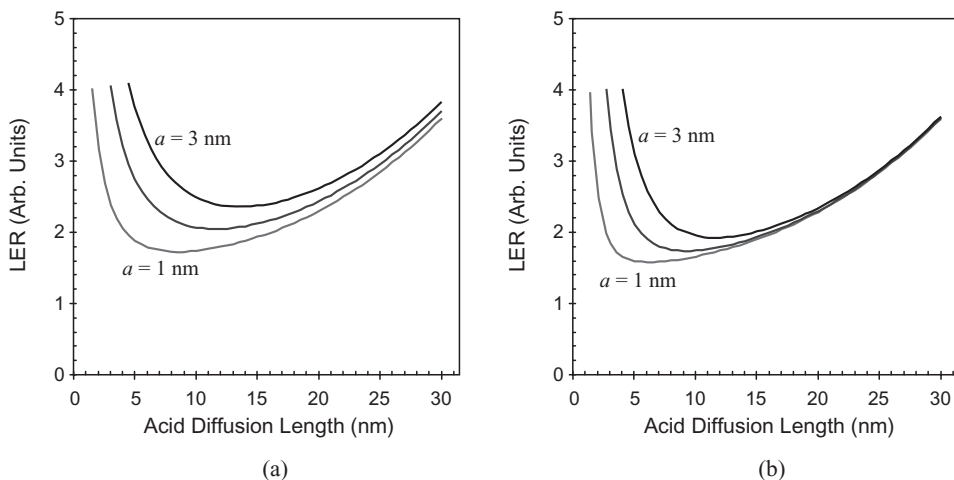


Figure 9.28 Prediction of LER trends for a 45-nm feature using the generic conditions found in Equation (9.99) and using three values of the deblocking reaction capture range a (1, 2 and 3 nm): (a) assuming a 2D problem, and (b) for a 3D problem

For $\sigma_D \ll L$ this becomes

$$LER \propto \left(1 + \frac{\pi^2 \sigma_D^2}{4L^2}\right) \sqrt{1 - (K_{\text{amp}} t_{\text{PEB}}) \langle m^* \rangle \ln \langle m^* \rangle \left(\frac{a}{\sigma_D}\right)^p \frac{\langle n_{0-\text{block}} \rangle}{\langle n_{0-\text{PAG}} \rangle}} \quad (9.98)$$

To evaluate this expression, consider some typical values. Let $M_0 N_A = 1.2/\text{nm}^3$, $G_0 N_A = 0.042/\text{nm}^3$, $\langle h \rangle = 0.3$, $\langle m^* \rangle = 0.6$. Thus, the LER becomes

$$LER \propto \frac{\sigma_D^2}{1 - e^{-\pi^2 \sigma_D^2 / 2L^2}} \sqrt{1 + 15 \left(\frac{a}{\sigma_D}\right)^p} \quad (9.99)$$

Figure 9.28 shows the trend of LER versus acid diffusion for a 45-nm feature for three different values of the deprotection capture range a , 1, 2 and 3 nm. In each case, there is a diffusion length that minimizes the LER. Below the optimum diffusion length, LER is limited by σ_{m^*} so that increasing the diffusion will improve LER. Above the optimum diffusion length the LER is gradient limited, so that increases in diffusion further degrade the gradient and worsen the LER.

9.9 Summary

The above discussion traces the aerial image log-slope through to its ultimate impact on exposure latitude (see Table 9.1). The resist contrast relates the original image contrast to the final development rate gradient, encompassing exposure, chemical amplification,

Table 9.1 Summary of lithography process steps and their corresponding information metrics

Process Step	Information	Error Sources	Information Metric
Design	Polygons, binary	(usually assumed perfect)	
Mask	Amplitude transmittance, $t_m(x,y)$	CD and registration errors, corner rounding, phase and transmittance	
Aerial Image	$I(x,y)$	Diffraction limitation, aberrations, defocus, flare, polarization	NILS
Image in Resist	$I(x,y,z)$	Substrate reflections/ thin film effects, polarization effects, defocus through the resist	NILS
Exposure	Latent Image $m(x,y,z)$ or $h(x,y,z)$ (before PEB)	Exposure dose errors	Latent image gradient
Post-exposure Bake	Latent Image $m^*(x,y,z)$ (after PEB)	Thermal dose errors, diffusion	Latent image gradient
Development	Development Rate $r(x,y,z)$ + Resist Profile (CD, sidewall angle, resist loss)	Finite contrast, $r_{\max}/r_{\min}$	Development rate log-slope, gamma + exposure latitude, CD error

diffusion and development. Since NILS degrades with focus in a predictable way, the final Equation (9.88) provides a simple means of capturing the impacts of both focus and exposure on final CD control. Obviously, this simple gradient approach has its limitations. In particular, the impact of isofocal bias on depth of focus is not captured by looking only at gradients. However, the gradient-based optimization embodied in the use of the normalized image log-slope has proven to be an extremely powerful theory-based guide to full lithographic optimization.

The chain rule of differentiation provides a simple approach for identifying the effects of each step in the process:

$$\frac{\partial \ln R}{\partial x} = \left(\frac{\partial \ln R}{\partial m^*} \right) \left(\frac{\partial m^*}{\partial m} \right) \left(\frac{\partial m}{\partial \ln E} \right) \left(\frac{\partial \ln I}{\partial x} \right) \quad (9.100)$$

Comparing this equation to the lithographic imaging Equation (9.53) shows that the photoresist contrast is a combined metric of exposure, PEB and development:

$$\gamma = \left(\frac{\partial \ln R}{\partial m^*} \right) \left(\frac{\partial m^*}{\partial m} \right) \left(\frac{\partial m}{\partial \ln E} \right) \quad (9.101)$$

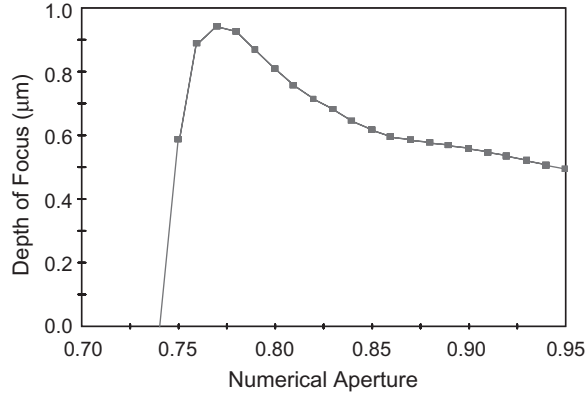
Problems

- 9.1. Using results from Chapter 3, show that the aerial image NILS falls off approximately quadratically with defocus.
- 9.2. Consider the three-beam aerial image of lines and spaces:

$$I(x) = a_0^2 + 4a_0a_1 \cos(2\pi x/p) + 4a_1^2 \cos^2(2\pi x/p)$$

Assuming equal lines and spaces, plot $p\text{ILS}(x)$ versus x/p for $0 \leq x \leq p/2$.

- 9.3. A plot of DOF versus NA (for a given feature) looks something like



Explain qualitatively why there is an optimum NA that maximizes the depth of focus.

- 9.4. Using the following approximation for exposure latitude,

$$EL = 20 \frac{\partial \ln E}{\partial \ln CD}$$

graphically compare Equations (9.6) and (9.91) [let $\gamma = 10$ and $D_{\text{eff}}/CD = 2$]. What are the best-fit values of α and β using NILS in the range from 1.5 to 3.5?

- 9.5. Using Equations (9.20) and (9.21),

a) Derive an equation for $\left. \frac{\partial m^*/\partial x}{\partial m/\partial x} \right|_{x=p/4}$ for $N = 4$.

b) For this case, derive an expression for the approximate magnitude of the relative error in Equation (9.22).

- 9.6. Derive Equation (9.44) [hint: first assume $m^* = \text{constant}$, then assume h_{eff} is constant].

- 9.7. Show that in the limit as $\eta \rightarrow 0$, Equation (9.39) becomes Equation (9.25).

- 9.8. For the blocked polymer latent image when quencher is used, show that the maximum relative LIG becomes

$$\left. \frac{\partial m^*/\partial x}{\text{ILS}} \right|_{\text{max}} \approx \frac{1}{\eta} (1 - q_0) \ln(1 - q_0)$$

whenever $\eta < \eta_c$.

- 9.9. Show that at the optimum conditions (i.e. when the blocked polymer LIG is at its maximum),

$$\frac{1 - e^{-\alpha_i \eta}}{\eta} = \frac{1}{h_{\text{eff}} + \eta}$$

- 9.10. Consider a resist obeying the original kinetic development model so that $n = 12$, $m_{\text{th}} = 0.65$, and $r_{\text{max}}/r_{\text{min}} = 2000$. If one desired to reformulate the resist to increase the maximum log-development rate versus m^* by 20 %, how much would each one of these variables have to be individually changed?
- 9.11. In the limit of $\sigma_D \gg a$, derive an approximate proportionality for LER. In this regime, how does increasing diffusion affect LER?

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